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Scalable Electronics for Modular Multi Layer e-Skin

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Abstract

This report measures communication, power and calibration in a four-module FSR prototype. Complete *FULL* and change-driven *DELTA* transmission were recorded at 200 Hz for all 15 combinations, with three zero-load windows each. Four-module *FULL* traffic was 6.747828 ± 0.001148 Mbit/s. Measured *FULL* packet lengths matched the protocol model at $N = 1\text{--}4$; mean USB rates differed from nominal 200 Hz predictions by $+0.029$ to $+0.125\%$. Its extension to $N = 100$ predicts 167.7184 Mbit/s but remains unvalidated. *DELTA* reduced same-rate packet bytes by 86.60–88.73% at zero load and 65.10–80.57% in five audited dynamic batches. The fitted zero-load ordinary-mask trend was $K_0(N) = 7.554189 + 0.582684N$, and the rolling stress line used $K = 254$. At 200 Hz, the model gives maximum module counts of 286/5 for *FULL*, 464/41 for zero-load *DELTA* and 501/10 for rolling- K under the 480 Mbit/s USB and 10 Mbit/s HOST SPI references (USB/HOST SPI); these are conditional planning values. Four-module branch current was 72.513 and 78.720 mA, and the lowest accepted module DMM minimum was 3.236 V. These static readings do not establish transient supply capacity. Separate 22-point sweeps calibrated both FSR layers of M1. In the FSR2 evaluation, FIT PRESS gave 10.22% mean layer-estimate APE and 171.69 g mean per-cell MAE across 21 validation captures, with a 40.12 g zero-load estimate. Dispersion was metric- and comparator-dependent. Within the reported dataset, measurements stop at four modules; transient supply behaviour, cross-module calibration and repeatability after reassembly were not measured.

Contents

Acknowledgements

1	Introduction	1
2	Background	2
2.1	Electronic skin and modular sensing	2
2.2	Why electronics scalability matters	2
2.2.1	Data scalability	2
2.2.2	Power and calibration scalability	2
2.3	Research gap and project position	3
2.4	Objectives and hypotheses	3
2.4.1	O1 — Scalable data architecture	3
2.4.2	O2 — Power scalability	3
2.4.3	O3 — FSR calibration scalability	4
3	Methods	5
3.1	Hardware design and physical structure	5
3.1.1	Module boundary, construction and FSR geometry	5
3.1.2	Local FSR readout	6
3.1.3	Layered structure and multi-module communication	6
3.2	Power scalability	8
3.2.1	Supply and measurements	8
3.2.2	Derived quantities and comparisons	9
3.3	Data scalability	9
3.3.1	Protocol and byte model	9
3.3.2	Experiment arrangement and outputs	10
3.4	Calibration scalability	11
3.4.1	Acquisition and calibration models	11
3.4.2	Validation and hypothesis tests	11
3.4.3	Scalability accounting	12

4	Results	13
4.1	Data scalability results	13
4.1.1	Zero-load traffic and byte accounting	13
4.1.2	Dynamic DELTA traffic	14
4.2	Power scalability	15
4.2.1	Electrical demand	15
4.2.2	Recorded voltage delivery	16
4.3	Calibration scalability	16
4.3.1	M1/FSR2 calibration response	16
4.3.2	M1/FSR2 load reconstruction	17
4.3.3	M1/FSR2 spatial dispersion	18
5	Discussion	20
5.1	Data scalability	20
5.2	Power scalability	20
5.3	Calibration scalability	21
5.4	Ethical considerations	21
5.5	System-level interpretation	22
6	Conclusion	23
A	Current hardware design records	24
A.1	Hardware renderings	24
A.2	Schematics and PCB layouts	25
B	Complete protocol definitions	31
B.1	Protocol stack	31
B.2	Host command (<i>DSCM</i>)	32
B.3	<i>ESK</i> frames: full synchronisation and <i>DELTA</i>	32
B.4	<i>MULI</i> and USB module blocks	33
C	Data scalability records	36
C.1	Measured and derived variables	36
C.1.1	Complete-frame fractions, coverage and averaging	37
C.2	Experiment arrangement	38
D	Power scalability: records and supplementary results	42
D.1	Data and reproducibility record	42
D.1.1	Continuous-read configurations	43
D.1.2	Historical operating-state comparator	44

D.1.3	Quality checks and unresolved metadata	45
D.2	Supplementary observations	47
E	Calibration records and second-layer evaluation	52
E.1	Current source records and acquisition scope	52
E.2	Independent application to M1/FSR1	53
E.2.1	FSR1 dispersion and model limitations	55
E.3	Reproduction and preservation	56

List of Figures

3.1	Layered construction of one sensing module.	5
3.2	Overall prototype hardware system.	7
3.3	Layered acquisition and communication path.	8
4.1	Transport-model validation and capacity projection.	14
4.2	Continuous-read electrical demand.	15
4.3	Recorded source and module voltages.	16
4.4	M1/FSR2 calibration model comparison.	17
4.5	M1/FSR2 load reconstruction error.	18
4.6	M1/FSR2 spatial dispersion.	19
A.1	Current hardware renderings.	24
A.2	Mainboard schematic.	25
A.3	Mainboard PCB layout.	26
A.4	FSR-layer schematic.	27
A.5	FSR-layer PCB layout.	28
A.6	ACC-layer schematic.	29
A.7	ACC-layer PCB layout.	30
C.1	DELTA packet-length examples.	40
D.1	Module voltage differences at each module count.	47
D.2	Single-module operating-state comparison.	48
D.3	Branch currents in continuous-read configurations.	49
D.4	Recorded temperature observations.	50
D.5	Current prediction from separate single-module references.	51
E.1	M1/FSR1 independent calibration model comparison.	53
E.2	M1/FSR1 four-method load error.	54
E.3	M1/FSR1 calibration dispersion.	55

List of Tables

4.1	Zero-load packet lengths, rates and USB traffic.	13
4.2	Dynamic DELTA performance.	14
4.3	Electrical demand across all module combinations.	15
4.4	M1/FSR2 four-method accuracy.	17
B.1	Protocol layers.	31
B.2	Host SPI command.	32
B.3	ESK header.	32
B.4	ESK frame layouts.	32
B.5	Mask-based <i>DELTA</i> payload.	33
B.6	<i>MULI</i> header.	34
B.7	<i>MULI</i> module block.	34
C.1	Variables used for the revised-firmware data analysis.	37
C.2	Complete zero-load matrix for each mode. Each subset has three consecutive windows.	38
C.3	Inventory and selection by recorded window. PASS/CHECK refers to main-window findings, not independent trial probabilities.	38
C.4	Dynamic batches on 7 September. Start time identifies the first capture of each three-window batch, using local recorded time.	39
D.1	All continuous-read combinations and branch-current sums.	43
D.2	Targeted module retests and retained historical values.	44
D.3	Historical blocked-standby record identifiers.	45
E.1	Current calibration source inventory.	52
E.2	M1/FSR1 four-method accuracy.	54

Chapter 1

Introduction

Electronic skin (e-skin) enables robots to sense contact over distributed surfaces. As the sensing area grows, however, the main challenge is no longer only the sensor material. Each additional sensing module increases the data that must be transferred, the electrical demand that must be supplied, and the calibration effort required to obtain comparable measurements.

This project focuses on the electronics scalability of a modular FSR-based e-skin developed from the existing modular prototype [DDZ⁺25]. The redesigned unit uses a hexagonal form, two replaceable FSR layers, and local acquisition on each module. The module data are then transferred through a shared Host SPI path and sent to the PC over USB. Consequently, the number of active modules can be treated as an explicit scalability variable.

The project addresses three practical questions. First, does complete raw-data transfer remain feasible as modules are added, and can a change-driven *DELTA* representation [Mis06] reduce the transmitted data volume when only a small part of the FSR state changes? Second, does the parallel regulated power arrangement maintain stable module operation as the system grows? Third, can a layer-wide FSR calibration reduce cell-to-cell variation and map ADC readings to applied pressure? These questions are evaluated using the implemented prototype, theoretical models, and clearly separated extrapolation beyond the measured module count.

The objectives and hypotheses are defined in Chapter 2. The hardware, power, data and calibration methods are then presented in Sections 3.1, 3.2, 3.3 and 3.4.

Chapter 2

Background

2.1 Electronic skin and modular sensing

Electronic skin aims to provide distributed contact information over a robotic or wearable surface. Flexible resistive arrays are attractive for this purpose because they can be arranged over a large area and read as a grid of individual sensing cells. However, increasing the number of cells also increases the number of measurements that must be acquired, transported and calibrated. Flexible tactile-array studies therefore treat the electrical readout and the repeatability of the array as part of the sensing system, not only as implementation details [FRJ⁺22].

The present project uses a modular FSR architecture to study this problem at system level. A module contains two replaceable FSR layers and performs local matrix acquisition before sending its data to a host. The hexagonal module shape follows the geometric motivation of honeycomb tiling: among equal-area partitions, regular hexagons minimise the total boundary length [Hal01]. The shape is therefore a design basis for arranging modules over a larger or curved surface; the scalability evaluation concerns the resulting electronics and data path.

2.2 Why electronics scalability matters

2.2.1 Data scalability

With frame-based transmission, every module sends its complete FSR state at every update, even when most cells have not changed. The resulting data volume grows with both module count and update rate. Send-on-Delta communication offers a general alternative: transmit a cell only when its value changes by a defined amount [Mis06]. Event-driven tactile encoding has also been used to reduce communication and latency [BMRD⁺17].

This project evaluates a narrower, reproducible form of that idea. The FSR arrays are still scanned synchronously; the local state is compared with the previous cached state, and the transport sends either a complete *FULL* frame or a change-driven *DELTA* frame [Mis06]. The comparison is therefore about data volume and reconstruction integrity, rather than a claim of fully asynchronous sensing.

2.2.2 Power and calibration scalability

Adding modules also increases the current drawn from the supply and can reduce voltage margin or increase temperature. A parallel regulated 3.3 V branch per module makes this dependence measurable through branch current and module voltage, while keeping the signal path separate from the power boundary.

The same modularity creates a calibration problem: FSR cells may differ in baseline, sensitivity and saturation behaviour, and replacement or reassembly can change those characteristics. Layer-wide loading and individual cell calibration are established ways to address this burden [KST⁺18]; in this project, the relevant question is whether the FSR calibration procedure remains repeatable as modules are added or serviced.

2.3 Research gap and project position

Much of the e-skin literature foregrounds sensing mechanisms, functional materials and local sensor performance. For large-area robotic e-skin, however, electronics, energy, information processing and transport introduce challenges beyond the sensing element itself [DYL⁺19]. The modular multi-layer system from which the present FSR prototype was developed provides a laboratory platform for robotics investigations [DDZ⁺25]. Moving such a prototype from the bench to broad deployment on a robot requires scalability to be treated as an explicit system-design problem.

A scalable architecture must therefore be assessed across the complete chain: local acquisition, module communication, host transport, power delivery and calibration. The planning documents identify data bandwidth, power stability and calibration repeatability as the main system-level risks. This report narrows that broad motivation to the current FSR prototype and evaluates data transfer, power stability and the current FSR calibration method. The results distinguish measured behaviour on the available modules from modelled larger-scale estimates.

2.4 Objectives and hypotheses

2.4.1 O1 — Scalable data architecture

Design and evaluate a data architecture that can scale from a single module to multiple modules. The architecture compares complete *FULL* transfer with change-driven *DELTA* transfer [Mis06], while retaining full stream reconstruction at the PC. A validated throughput model is also intended to support estimates for larger module configurations.

D-H1 within the one-to-four-module range, observed mean USB packet lengths follow the *FULL* byte budget and the *DELTA* budget parameterised by recorded mask activity, complete-frame incidence and module-update coverage. Predictions beyond this range are conditional on specified inputs and are assessed separately from delivery integrity;

D-H2 the change-driven, event-inspired *DELTA* strategy [Mis06] reduces same-rate packet bytes under the tested activity conditions within windows with no detected sequence, cache-chain or target-update anomalies.

2.4.2 O2 — Power scalability

Characterise how electrical demand changes with module count and whether the parallel 3.3 V supply delivers voltage within the specified band; distinguish these readings from a verified capacity limit.

P-H1 currents measured for modules operating separately provide a useful prediction of the demand of their tested multi-module combinations under comparable operating conditions; prediction errors are evaluated descriptively, without treating a current-versus-count fit as proof;

P-H2 recorded module voltages remain within the project-defined $3.3\text{ V} \pm 5\%$ band in the tested configurations. Transient voltage delivery and fault-free operation require additional verified monitoring.

2.4.3 O3 — FSR calibration scalability

Develop a layer-wide acquisition workflow that produces individual cell mappings from ADC readings to equivalent whole-layer calibration mass. Evaluate its application to separately calibrated FSR layers and compare prediction error and spatial dispersion under matched loading and analysis conditions. Apply the same acquisition, fitting and validation procedure independently to a second nominally identical FSR layer to test whether its reconstruction benefit is retained beyond the primary layer. Cross-module transfer and retention after reassembly remain validation targets.

C-H1 with matched RAW inputs and common valid cells, per-cell multi-load fitting can reduce spatial dispersion relative to two-point LINEAR or GAMMA calibration. The three outputs are compared in g using CV, variance, IQR and MAD; the result is evaluated separately for each metric and load, without comparing ADC dispersion directly with mass dispersion;

C-H2 per-cell calibration provides an estimate of known applied whole-layer mass. Validation quantifies signed bias, absolute error, per-cell MAE/RMSE and non-zero-load percentage error over the measured range. No prespecified accuracy acceptance threshold is documented, so these are descriptive error assessments rather than a pass/fail demonstration of acceptable accuracy.

C-H3 when the same calibration workflow is applied independently to a second nominally identical FSR layer, FIT PRESS will reduce non-zero-load APE and per-cell MAE/RMSE relative to two-point LINEAR and GAMMA calibration under the same validation procedure. This tests repeatability of the method across the two available layers, rather than direct transfer of one fitted model without retraining.

Chapter 3

Methods

3.1 Hardware design and physical structure

3.1.1 Module boundary, construction and FSR geometry

The evaluated unit is a repeatable sensing module with a rigid hexagonal mainboard and two replaceable flexible FSR layers.

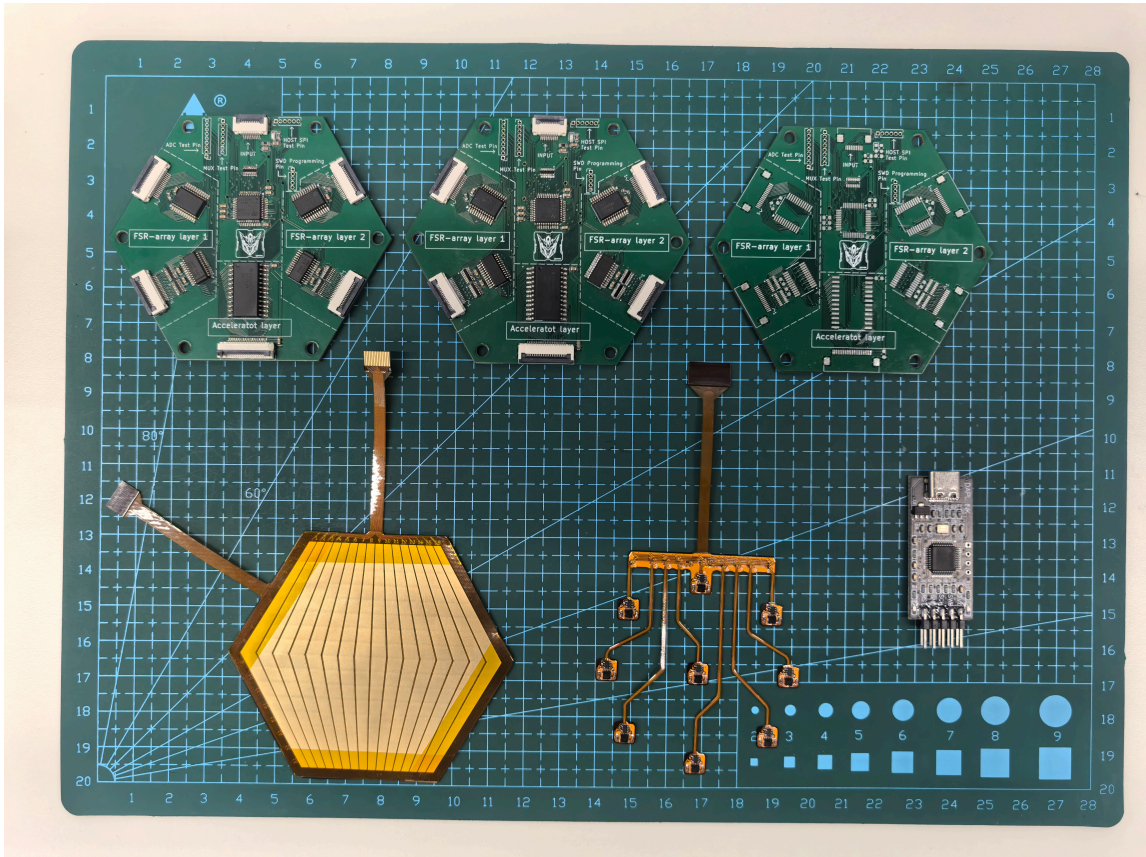


Figure 3.1: Layered construction of one sensing module.

This choice follows the honeycomb theorem: for a partition into equal-area regions, the regular hexagonal tiling has the minimum total perimeter [Hal01]. Thus, for a given module area, a hexagonal layout reduces shared boundary length relative to a square tiling while preserving a regular repeatable pattern. This is the

geometric design rationale; mechanical and electrical benefits are not inferred from the theorem and require separate measurements. The current renderings, schematics and PCB layouts are provided in Appendix A. This report focuses on FSR sensing; the ACC currently exists only as a hardware prototype.

The mainboard source is a four-layer rigid FR4 PCB. It carries the local STM32G474CETx microcontroller, the FSR readout electronics, power distribution, programming access, test points, and connectors for the replaceable sensing layers. The FSR-array electrode carrier is a two-layer flexible PCB. During assembly, a pressure-sensitive resistive foam—the piezoresistive sensing layer—is sandwiched between the opposing copper electrode surfaces [DDZ⁺25]. One surface carries 16 row electrodes and the other carries 16 perpendicular column electrodes. Compression changes the foam resistance locally, so each row–column intersection forms one cell of the 16×16 matrix. The intermediate material is therefore a resistive sensing layer rather than a conventional insulating spacer. FSR1 and FSR2 are separate assemblies with separate row and column interfaces on the mainboard, so either layer can be removed or replaced without redesigning the other.

3.1.2 Local FSR readout

Each module acquires its FSR layers locally. The STM32 selects one row through the multiplexer, samples the column voltages with the ADC, and repeats this for the complete matrix. The two matrices are buffered locally before the module frame is assembled, so adding a module adds an independent acquisition node rather than extending the host’s sensor inputs.

3.1.3 Layered structure and multi-module communication

The system has three functional layers. Each module acquires its two FSR arrays locally, and its STM32 produces a complete or change-based module representation. The modules share the Host SPI clock and data lines while module selection remains separate. The current integration and measurements use up to four modules. Here N denotes active modules and M the configured maximum number of protocol slots, with $N \leq M$ and $M = 4$ in this project. Larger configurations are theoretical extensions requiring corresponding host and protocol changes.

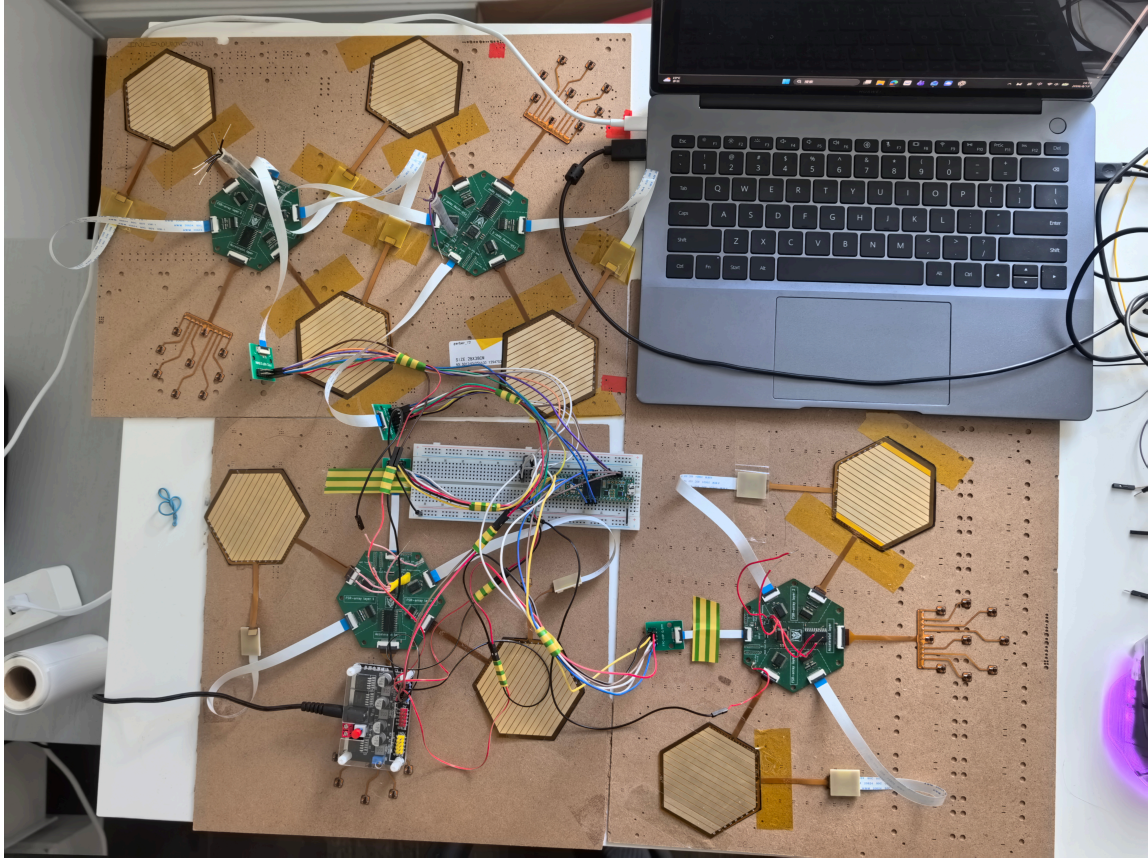


Figure 3.2: Overall prototype hardware system.

The Teensy¹ is the Host SPI master: it selects each module, clocks only its complete encoded *ESKF* or *ESKD* frame, including its header and CRC but no tail padding, and combines the returned module data into a *MULI* packet. The PC receives the packet over USB and reconstructs the FSR state.

¹The Teensy serves as the host controller in the present prototype; it is not an architectural requirement. A higher-performance MCU or FPGA providing a compatible SPI interface could replace it.

E-SKIN hardware hierarchy and transmitted fields

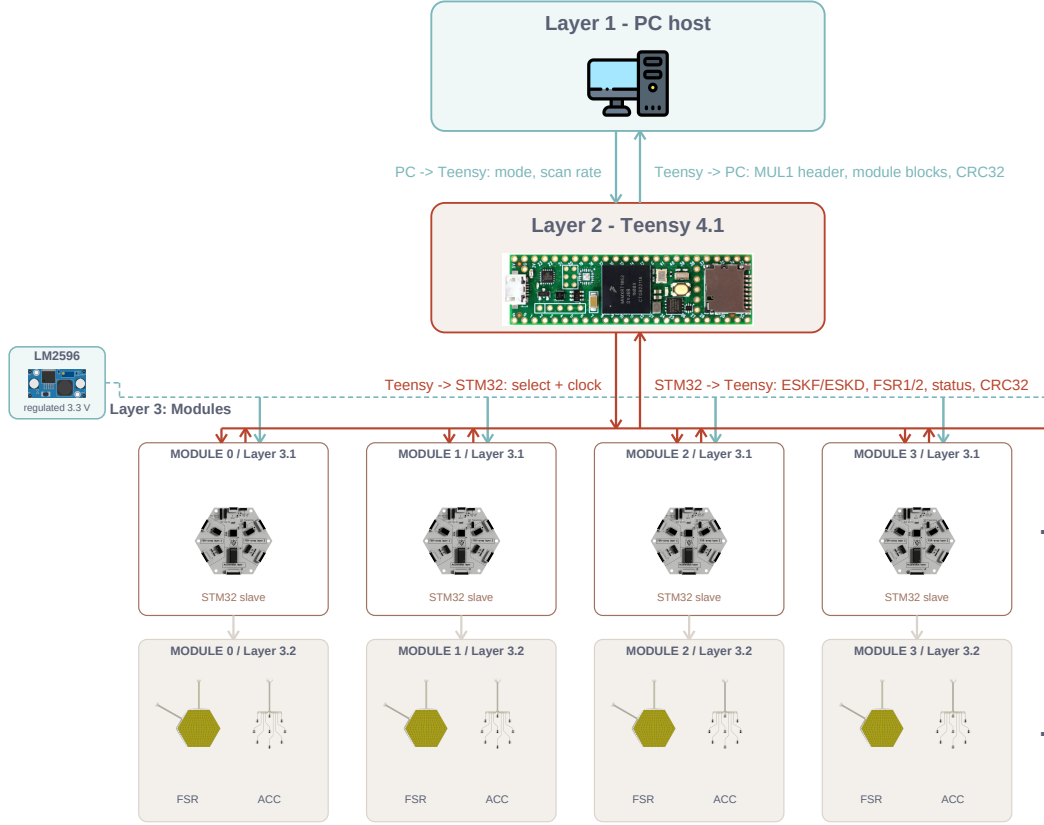


Figure 3.3: Layered acquisition and communication path.

An LM2596 step-down regulator [Tex23] supplied a common external 3.3 V bus. Each module was connected through a separate branch with a common return. The supply boundary and the voltage, current and temperature measurements are described in Section 3.2.

The acquisition sequence is therefore: FSR matrix $\rightarrow$ local STM32 buffer $\rightarrow$ Host SPI *ESK* frame $\rightarrow$ Teensy *MUL1* aggregation $\rightarrow$ USB serial $\rightarrow$ PC parser. The detailed packet model is defined in Section 3.3.1.

3.2 Power scalability

3.2.1 Supply and measurements

An LM2596 regulator supplied a common 3.3 V bus with one branch per module and a common return (Figure 3.3). The Teensy was powered separately by USB and excluded from the measurement boundary. Tests used *FULL* transport, continuous host reading and a configured 200 Hz scan rate.

All 15 non-empty combinations of M0–M3 were recorded without applied load (ZERO) and under a full-coverage high-load condition labelled MAX, which was used to probe a representative adverse operating case. This gave 30 configuration entries and 64 selected branch observations. Seven targeted branch updates are included in this compiled matrix but do not form complete configuration repeats. Their source records are listed in Appendix D.

3.2.2 Derived quantities and comparisons

Source voltage, module voltage and branch current were read sequentially with an FNIRSI 2C53P DMM. For a module set S ,

$$I_{\Sigma}(S) = \sum_{i \in S} I_i, \quad \hat{P}_{\text{out}}(S) = V_s I_{\Sigma}(S), \quad (3.1)$$

$$\Delta V_i = V_s - V_i, \quad H_i = \min(V_{i,\text{min}}) - 3.135 \text{ V}. \quad (3.2)$$

P-H1 was tested by comparing each multi-module current with the sum of the corresponding modules measured separately under the same recorded load label,

$$e_I(S) = 100 \frac{I_{\Sigma}(S) - \sum_{i \in S} I_i^{N=1}}{\sum_{i \in S} I_i^{N=1}} \%. \quad (3.3)$$

These comparisons share the four single-module references and assume that the recorded module assignments were retained. P-H2 compared recorded module voltages with the project band of 3.135–3.465 V. The measurements describe static operating points; they do not measure startup peaks, ripple, load-switching transients, thermal equilibrium or usable supply capacity.

3.3 Data scalability

Each STM32 scans both FSR matrices on every commanded round. In *FULL* mode it transmits the complete state; in *DELTA* mode it transmits only cells whose change from the cached value exceeds eight ADC codes. The PC applies these masked updates to a separate cache for each module. A complete *ESKF* frame initialises or restores the cache and is also emitted by the one-second synchronisation policy. Sparsity is therefore introduced after synchronous acquisition, following the send-on-delta principle [Mis06], rather than by an asynchronous sensor.

3.3.1 Protocol and byte model

The STM32 sends either a 1044 B complete *ESKF* frame or a masked *ESKD* frame of length $84 + 2K$ B. Here K is the number of cells whose ADC values meet the configured change criterion relative to the cached state and are therefore transmitted in that *DELTA* update. The Teensy combines returned frames in a *MULI* USB packet. The complete fields are given in Appendix B. With N active modules, M the maximum module count configured in the current system (equivalently, the number of reserved protocol slots), and round frequency f , the two simple packet-rate budgets are

$$R_{\text{USB},\text{FULL}} = \frac{8f[24 + 4M + 1044N]}{10^6}, \quad (3.4)$$

$$R_{\text{USB},\text{ESKD}} = \frac{8f[24 + 4M + N(84 + 2K)]}{10^6} \text{ Mbit/s}. \quad (3.5)$$

The prototype has four fixed slots, so its USB overhead is $24 + 4M = 40$ B for every packet. Projections beyond four modules instead use $M = \max(4, N)$ and require corresponding protocol and scheduling extensions.

For the capacity projection, the measured zero-load ordinary-mask means at $N = 1\text{--}4$ were fitted with an

unweighted linear model,

$$K_0(N) = 7.554189 + 0.582684N, \quad (3.6)$$

with $R^2 = 0.9519$. A second DELTA stress line held $K = 254$, the largest ordinary K observed across the three rolling windows (Appendix C). Both projections used the measured one-second synchronisation policy as the reference fraction $q = 0.005$ at 200 Hz. The corresponding mean encoded bytes per active module were

$$\ell_D(K) = (1 - q)(84 + 2K) + q(1044), \quad (3.7)$$

so that the aggregate HOST SPI payload and USB application demand were $8fN\ell_D(K)/10^6$ and $8f[24 + 4\max(4, N) + N\ell_D(K)]/10^6$ Mbit/s, respectively. The comparison uses 480 Mbit/s as the theoretical USB reference and 10 Mbit/s as the raw 10 MHz HOST SPI reference; these are planning ceilings rather than measured achieved capacities.

For module i , let q_i be the observed fraction of valid inner frames that are complete and let $\bar{K}_{D,i}$ be the mean mask population among its ordinary *ESKD* frames. Its mean encoded length is

$$\bar{\ell}_i = 1044q_i + (1 - q_i)(84 + 2\bar{K}_{D,i}). \quad (3.8)$$

Thus periodic and other observed complete frames are included rather than added as an independent stream. For P valid outer packets, update coverage c_i and other retained slot bytes B_{other} , the corresponding mean packet length is

$$\bar{L}_{\text{USB}} = 24 + 4M + \sum_{i=1}^M c_i \bar{\ell}_i + \frac{B_{\text{other}}}{P}. \quad (3.9)$$

The same quantity was checked byte by byte using

$$B = P(24 + 4M) + 84n_D + 2 \sum_{d=1}^{n_D} K_d + 1044n_F + B_{\text{other}}, \quad (3.10)$$

where n_D and n_F count valid ordinary and complete inner frames. These relations account for received bytes; delivery integrity is evaluated separately.

3.3.2 Experiment arrangement and outputs

The final evaluated implementation used the variable-length *FULL/DELTA* firmware at a configured 200 Hz. A batch comprised three consecutive recordings of approximately 40 s under one configuration. The fixed main window ran from 10 s to the end of each recording; the first 10 s remained available for startup and integrity checks. The three windows are temporal repeats within one setup.

The zero-load experiment covered all 15 non-empty subsets of four slots in both modes, giving 45 selected *FULL* and 45 selected *DELTA* windows. Seven three-window dynamic *DELTA* batches covered three action classes: large-area loading of M1 (including press/release), rolling input on M1, and simultaneous full-coverage loading of multiple modules. Five passed the main-window audit and supplied 15 performance windows. Additional experiments affected by defects in earlier protocol versions or by hardware faults were retained as diagnostic records and excluded from the performance summaries. The complete inventory and selection are listed in Appendix C.2.

The audit checked CRC and layout, outer-packet sequence, per-slot sequence, target-slot coverage and application of each update to an anchored cache. An *ESKF* establishes a new cache base but does not erase an earlier

sequence discontinuity. For a main window of duration T , retained byte volume B and count P of complete, valid *MULI* packets, the reported quantities are

$$f_{\text{eff}} = \frac{P}{T}, \quad \bar{L}_{\text{USB}} = \frac{B}{P}, \quad R_{\text{USB,meas}} = \frac{8B}{10^6 T}. \quad (3.11)$$

Same-rate saving is calculated from packet length,

$$\text{Reduction} = 100 \left(1 - \frac{\bar{L}_{\text{USB,DELTA}}}{24 + 4M + 1044N} \right) \%. \quad (3.12)$$

This prevents packet-rate variation from appearing as compression. Means and sample SDs use window-level values; n denotes windows, not frames or independent hardware assemblies.

3.4 Calibration scalability

Calibration was performed separately for the two FSR layers of M1. FSR2 is the main evaluation and FSR1 is reported in Appendix E.2; no model was transferred between layers.

3.4.1 Acquisition and calibration models

Each layer used a 22-point whole-layer loading sweep from 0 to 5000 g. At each mass, 200 fresh *FULL* frames were recorded and reduced to a temporal median for every taxel. Loading pieces were centred, and a silicone pad together with rigid support boards was used to distribute the applied load as evenly as practicable across the sensing area. However, the force at each location was not measured. Outputs are therefore equivalent whole-layer calibration mass, not validated local pressure.

Four mappings were constructed from the same sweep. LINEAR used each cell’s measured zero and 5000 g endpoints. GAMMA applied a fixed $\gamma = 4$ transform between those endpoints. FIT PRESS used isotonic regression followed by an inverse lookup table for each cell; equal ADC knots were merged and assigned the midpoint of their load interval. Shared LUT applied the same nonlinear procedure to the layer-mean ADC and then used that single inverse for every cell. For cell i , the four estimates can be written

$$\hat{m}_i \in \{f_{i,\text{LIN}}(a_i), f_{i,\gamma}(a_i), f_{i,\text{FIT}}(a_i), f_{\text{shared}}(a_i)\}. \quad (3.13)$$

3.4.2 Validation and hypothesis tests

FSR2 validation contained 21 complete captures at 20 distinct loads from 0 to 1969 g. Training models were frozen before these records were evaluated, and all comparisons used the same 256 cells and the temporal-median ADC matrix from each capture.

C-H1 compared the spatial dispersion of LINEAR, GAMMA and FIT PRESS in grams using CV, population variance, IQR and MAD. C-H2 evaluated reconstruction of the applied mass using the layer mean $\bar{m} = |C|^{-1} \sum_{i \in C} \hat{m}_i$, signed bias $b = \bar{m} - m$, mean APE for $m > 0$, and per-cell MAE and RMSE. Percentage error is undefined at zero, where bias is reported in grams. Shared LUT was retained as an accuracy comparator but was not part of the C-H1 dispersion test.

Here n is the number of saved captures at a load. Error bars show sample SD only where $n \geq 2$; the 200 frames and 256 spatial cells are not independent reapplications of the load.

3.4.3 Scalability accounting

For K training masses and N two-layer modules, layer-wide acquisition requires $2NK$ matrix captures and produces $512N$ cell models. Loading every cell separately at the same references would require $512NK$ captures. The resulting 256-fold ratio is a capture-count comparison; setup, settling, validation and processing time are additional costs.

Chapter 4

Results

4.1 Data scalability results

4.1.1 Zero-load traffic and byte accounting

The final experiment inventory contains 114 recordings: 48 *FULL* zero-load, 45 *DELTA* zero-load and 21 dynamic *DELTA*. The selected performance set contains 90 zero-load windows and 15 dynamic windows.

Table 4.1: Zero-load results, mean $\pm$ sample SD across fixed main windows. Each combination contributes three windows; n is the total number of windows at each N .

Mode	N	n	Mean packet (B)	Packet rate (Hz)	USB (Mbit/s)
<i>FULL</i>	1	12	1,084	200.251 ± 0.118	1.736573 ± 0.001025
<i>DELTA</i>	1	12	145.24 ± 16.50	200.158 ± 0.278	0.232562 ± 0.026405
<i>FULL</i>	2	18	2,128	200.187 ± 0.111	3.407991 ± 0.001895
<i>DELTA</i>	2	18	251.98 ± 16.36	200.373 ± 0.946	0.403900 ± 0.025933
<i>FULL</i>	3	12	3,172	200.059 ± 0.302	5.076696 ± 0.007657
<i>DELTA</i>	3	12	360.74 ± 24.15	200.159 ± 0.053	0.577646 ± 0.038721
<i>FULL</i>	4	3	4,216	200.066 ± 0.034	6.747828 ± 0.001148
<i>DELTA</i>	4	3	475.15 ± 5.72	200.600 ± 0.845	0.762543 ± 0.012375

The model predicted packet lengths of 1084, 2128, 3172 and 4216 B for $N = 1-4$, matching every measured *FULL* mean with 0 B error. Relative to the nominal 200 Hz prediction, measured mean USB rates differed by +0.125, +0.094, +0.029 and +0.033%, respectively. At $N = 100$ the extrapolated model gives 104,824 B per packet and 167.7184 Mbit/s. Zero-load *DELTA* reduced same-rate packet bytes by 86.60, 88.16, 88.63 and 88.73% for $N = 1-4$.

The zero-load ordinary-mask means were 8.26, 8.64, 9.10 and 10.04 cells per active module for $N = 1-4$; their fitted extension was $K_0(N) = 7.554189 + 0.582684N$ ($R^2 = 0.9519$). The rolling stress line used the maximum ordinary value $K = 254$ observed across its three windows. At 200 Hz, the continuous intersections with the 480 Mbit/s USB and 10 Mbit/s HOST SPI references were respectively $N = 286.24/5.99$ for *FULL*, $N = 464.25/41.22$ for the zero-load K_0 projection, and $N = 501.41/10.52$ for the rolling- K projection. The corresponding largest integer counts were 286/5, 464/41 and 501/10.

Mean ordinary *ESKD* mask population per active module was 8.26, 8.64, 9.10 and 10.04 for $N = 1-4$. The 45 zero-load *DELTA* windows contained 576,926 valid module frames, of which 2,884 (0.500%) were *ESKF*. For the 10 s-to-end analysis interval of each of the 114 recorded runs, the byte total reconstructed using Equation 3.10 exactly matched the recorded byte total; recorded minus reconstructed bytes was 0 B.

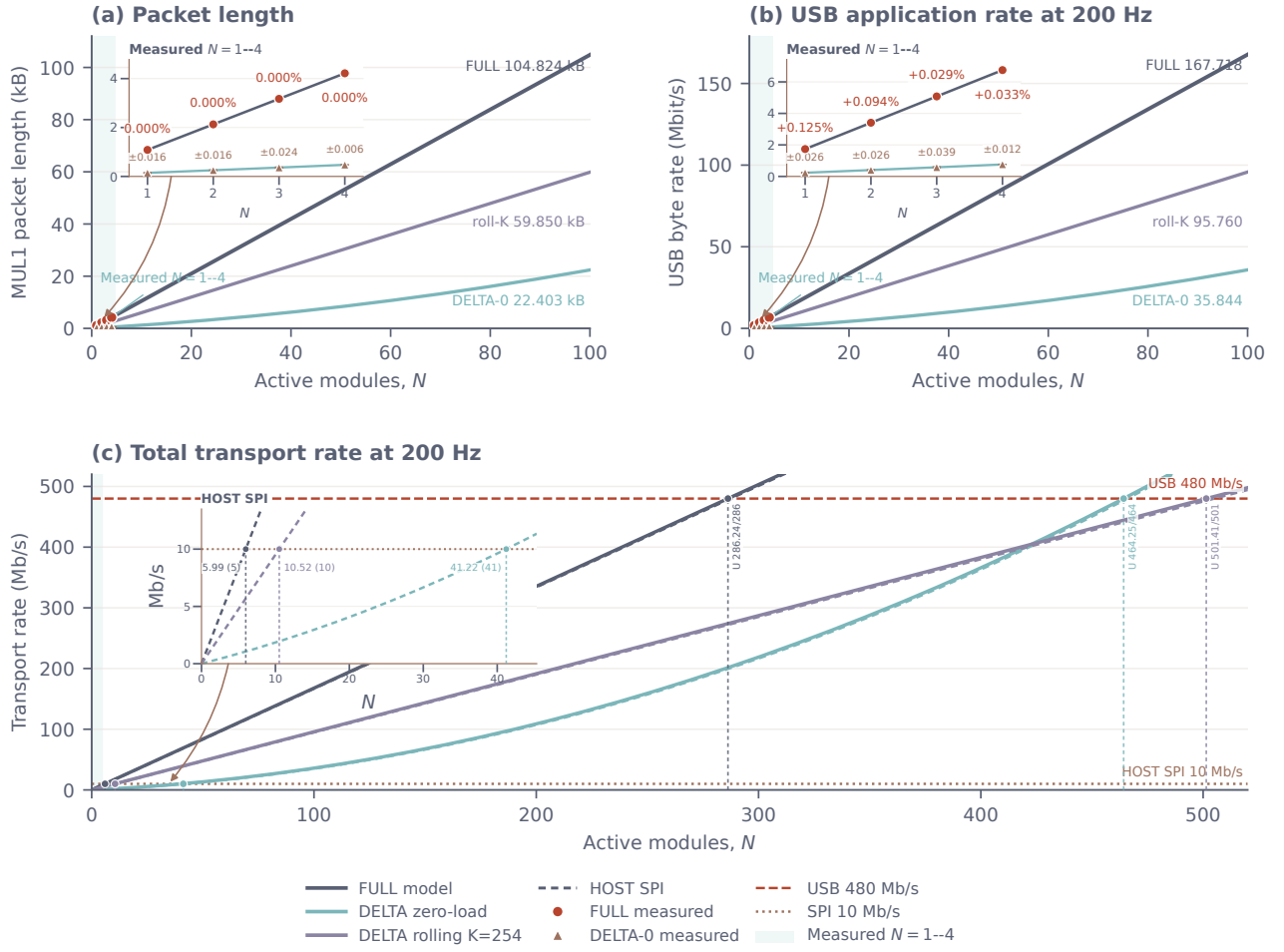


Figure 4.1: (a) FULL and zero-load DELTA packet lengths, (b) their USB application rates at 200 Hz, and (c) USB and HOST SPI transport rates in Mb/s. FULL points and error bars are window means $\pm$ sample SD; DELTA triangles show the corresponding zero-load means. The zero-load DELTA line uses the fitted $K_0(N)$ trend, while the rolling line uses the maximum ordinary $K = 254$ observed in the rolling batch. The lower panel includes 480 Mbit/s USB and 10 Mbit/s raw HOST SPI references; markers identify continuous intersections and the printed values give the largest integer module count at each reference. All $N > 4$ segments are conditional protocol-model extrapolations.

4.1.2 Dynamic DELTA traffic

Table 4.2: Dynamic results, mean $\pm$ sample SD across three consecutive windows. Reduction is relative to *FULL* packet length at the same module count.

Input / slots	PASS/ n	Mean packet (B)	Rate (Hz)	Reduction (%)
M1 large area	3/3	282.42 $\pm$ 4.74	200.080 $\pm$ 0.036	73.95 $\pm$ 0.44
M1 press/release	3/3	210.64 $\pm$ 10.57	200.422 $\pm$ 0.045	80.57 $\pm$ 0.98
M1 rolling	3/3	357.72 $\pm$ 11.18	200.356 $\pm$ 0.036	67.00 $\pm$ 1.03
M1+M3 simultaneous	3/3	510.90 $\pm$ 11.14	199.617 $\pm$ 0.015	75.99 $\pm$ 0.52
M0–M3 simultaneous	3/3	1471.55 $\pm$ 29.89	199.414 $\pm$ 0.068	65.10 $\pm$ 0.71

The five passing batches produced same-rate reductions from 65.10 ± 0.71 to $80.57 \pm 0.98\%$.

4.2 Power scalability

4.2.1 Electrical demand

All 15 module combinations were recorded under both load labels. At $N = 4$, branch-current sum was 72.513 mA for ZERO and 78.720 mA for MAX; the corresponding output-side estimates were 237.408 and 257.651 mW. Each $N = 4$ condition has one configuration entry.

Table 4.3: Mean $\pm$ sample SD across n_c module combinations. Each exact combination has one assembled entry; SD is unavailable at $N = 4$.

N	n_c	I_Σ (mA)		$\hat{P}_{\text{out}}$ (mW)	
		ZERO	MAX	ZERO	MAX
1	4	18.290 ± 0.182	20.006 ± 0.285	59.999 ± 0.602	65.633 ± 0.927
2	6	36.253 ± 0.276	39.629 ± 0.315	118.872 ± 0.885	129.891 ± 1.020
3	4	54.426 ± 0.346	58.849 ± 0.482	178.313 ± 1.119	192.761 ± 1.612
4	1	72.513	78.720	237.408	257.651

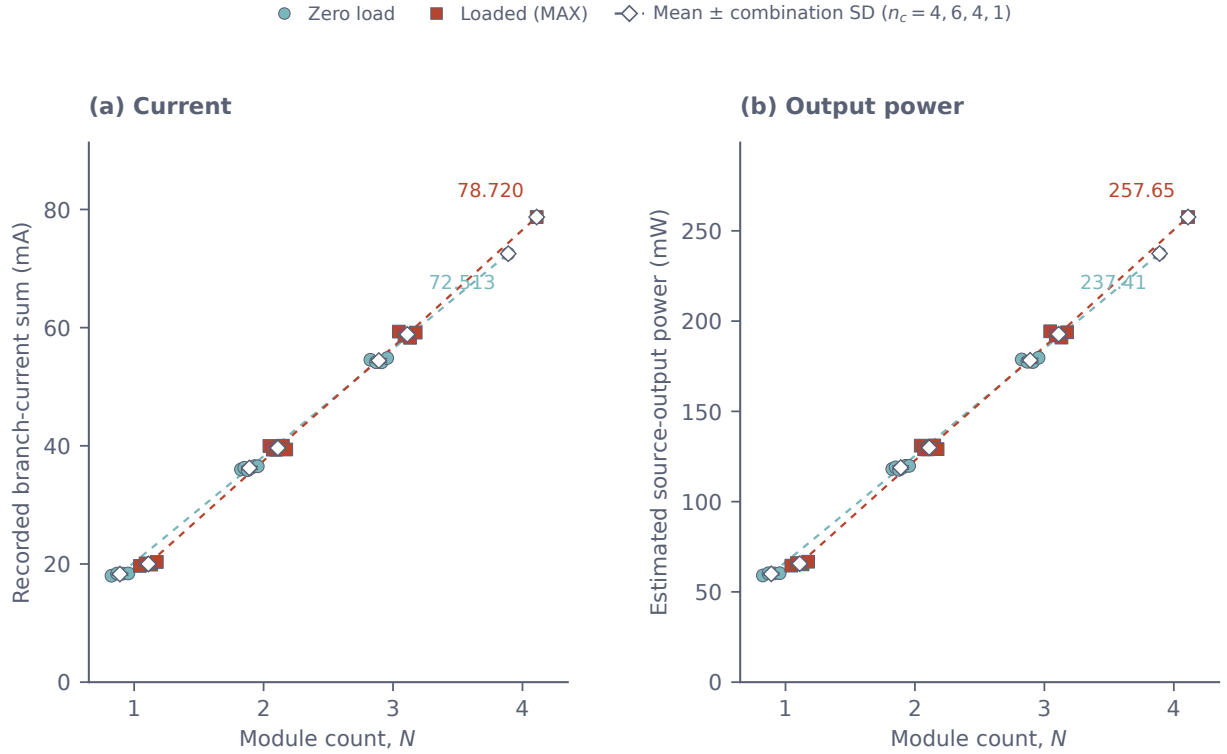


Figure 4.2: Current and output-side power estimates for all assembled configurations, classified by recorded load type. Diamonds and error bars show the mean $\pm$ sample SD across combinations; $N = 4$ has $n_c = 1$.

Across the 15 matched combinations, MAX current exceeded ZERO by 7.66–11.03%. For the eleven multi-module combinations, residuals against the summed $N = 1$ references were -1.483 to -0.256% for ZERO and -2.726 to $+0.460\%$ for MAX. Individual comparisons are shown in Appendix Figure D.5.

4.2.2 Recorded voltage delivery

All accepted module readings from the 30 configuration entries lay within 3.135–3.465 V. The lowest recorded DMM MIN values were 3.240 V for ZERO and 3.236 V for MAX, 105 and 101 mV above the lower limit.

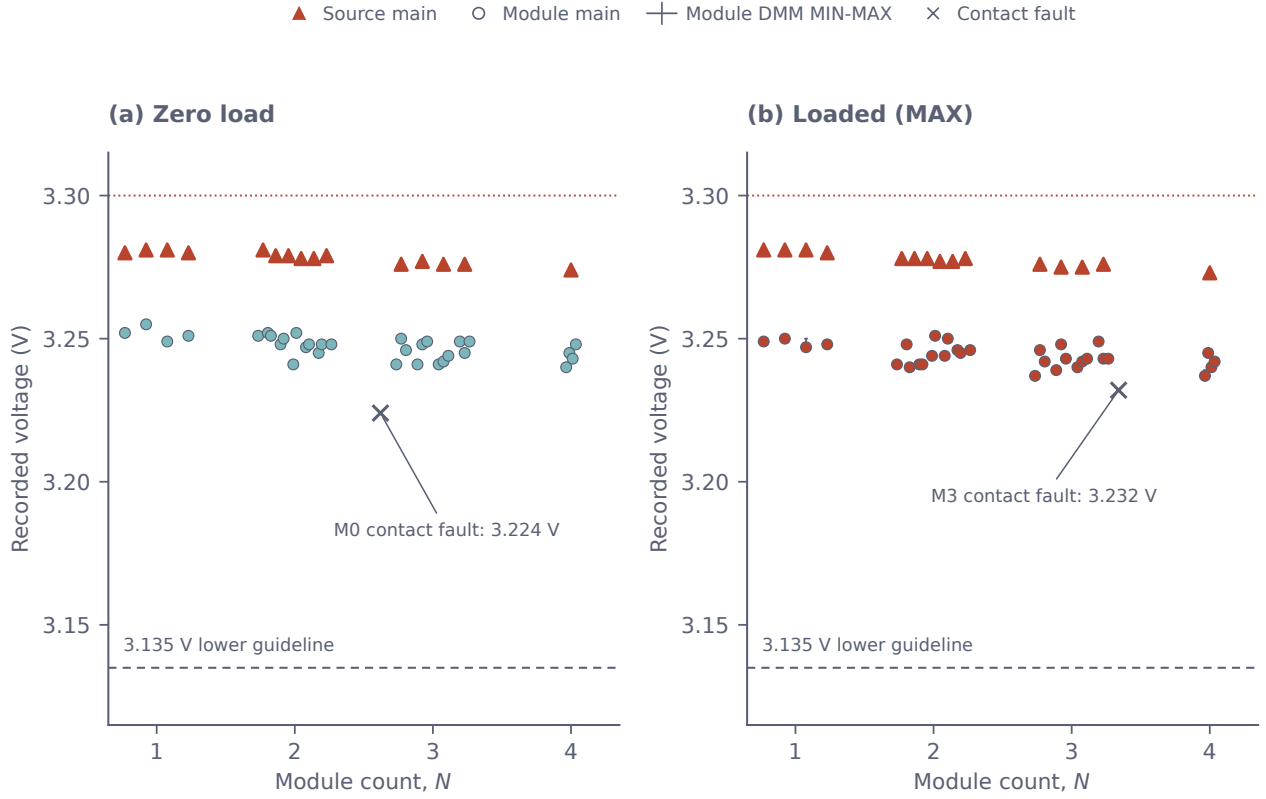


Figure 4.3: Main readings and DMM MIN–MAX segments for all combinations. Markers distinguish source main readings from module main readings; crosses identify two recorded contact-fault readings.

The accepted source-to-module differences were 26–38 mV for ZERO and 26–39 mV for MAX. Across 32 matched module observations, the loaded-minus-zero difference ranged from -4 to $+10$ mV: 21 were positive, six negative and five zero. The 3.224 V M0 and 3.232 V M3 contact-fault records are retained in Appendix D.

4.3 Calibration scalability

4.3.1 M1/FSR2 calibration response

FSR2 used 22 training masses from 0 to 5000 g and 200 *FULL* frames per mass. All 256 cells produced an inverse LUT. Full-minus-zero span ranged from 729 to 1438 ADC codes. One zero-load and 20 upper-end captures exceeded the predefined 50-code within-capture range flag.

Plateau merging affected 138 of 256 inverse tables and removed 268 duplicate knots across 170 flat load intervals. Median interval width was 73 g and the maximum was 3855 g; 90 tables had a plateau touching zero and three had one touching 5000 g.

M1 / FSR2: calibration functions ($\gamma = 4$)

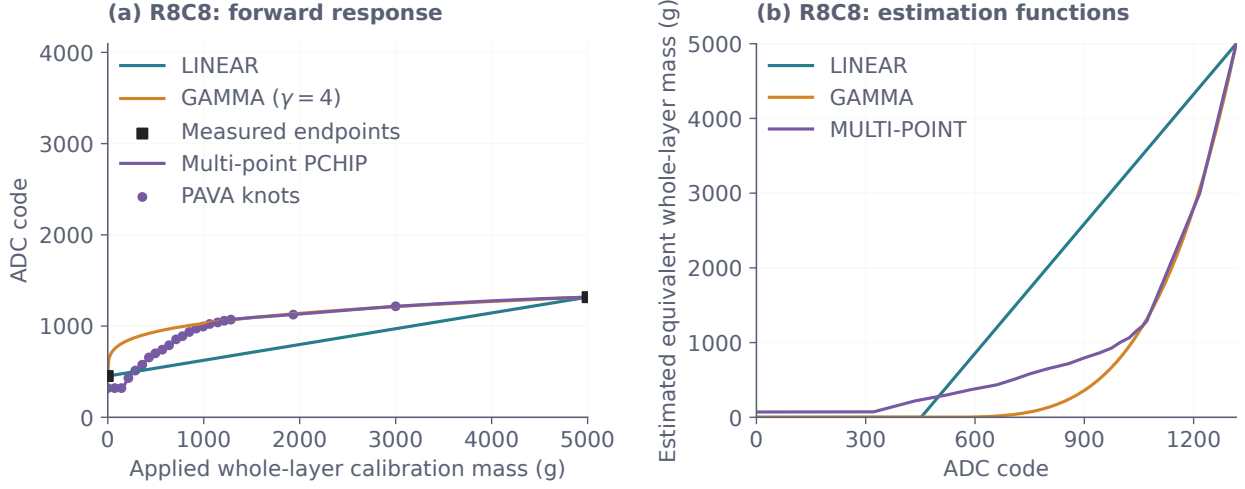


Figure 4.4: Forward and inverse mappings for representative cell R8C8. Each training point is the median of 200 fresh frames from one capture at that mass.

4.3.2 M1/FSR2 load reconstruction

Validation contained 21 captures at 20 loads from 0 to 1969 g. The 140 g load had $n = 2$ captures; every other load had $n = 1$. All methods used the same 256-cell population.

Table 4.4: Capture-weighted accuracy on 256 cells. Mean APE uses 20 non-zero captures; gram-valued summaries use all 21 captures.

Method	Mean APE (%)	Mean $ b $ (g)	Mean cell MAE (g)	Mean cell RMSE (g)
Linear	188.35	1205.92	1241.30	1526.25
Gamma ($\gamma = 4$)	35.75	115.59	586.01	735.70
Per-cell LUT	10.22	47.24	171.69	312.42
Shared LUT	33.01	275.67	638.14	943.52

FIT PRESS gave a mean layer-estimate APE of 10.22%, compared with 188.35% for LINEAR, 35.75% for GAMMA and 33.01% for Shared LUT. Its mean absolute layer bias was 47.24 g and mean per-cell MAE was 171.69 g. Equal weighting of distinct non-zero masses gave a FIT PRESS mean APE of 9.61%.

At zero load, LINEAR, GAMMA, FIT PRESS and Shared LUT reconstructed 12.61, 0.04, 40.12 and 35.98 g. At 1969 g, FIT PRESS reconstructed 2015.43 g; its spatial SD was 684.23 g and per-cell MAE was 505.70 g. At 140 g, the two FIT PRESS capture means were summarised as 170.55 ± 39.70 g.

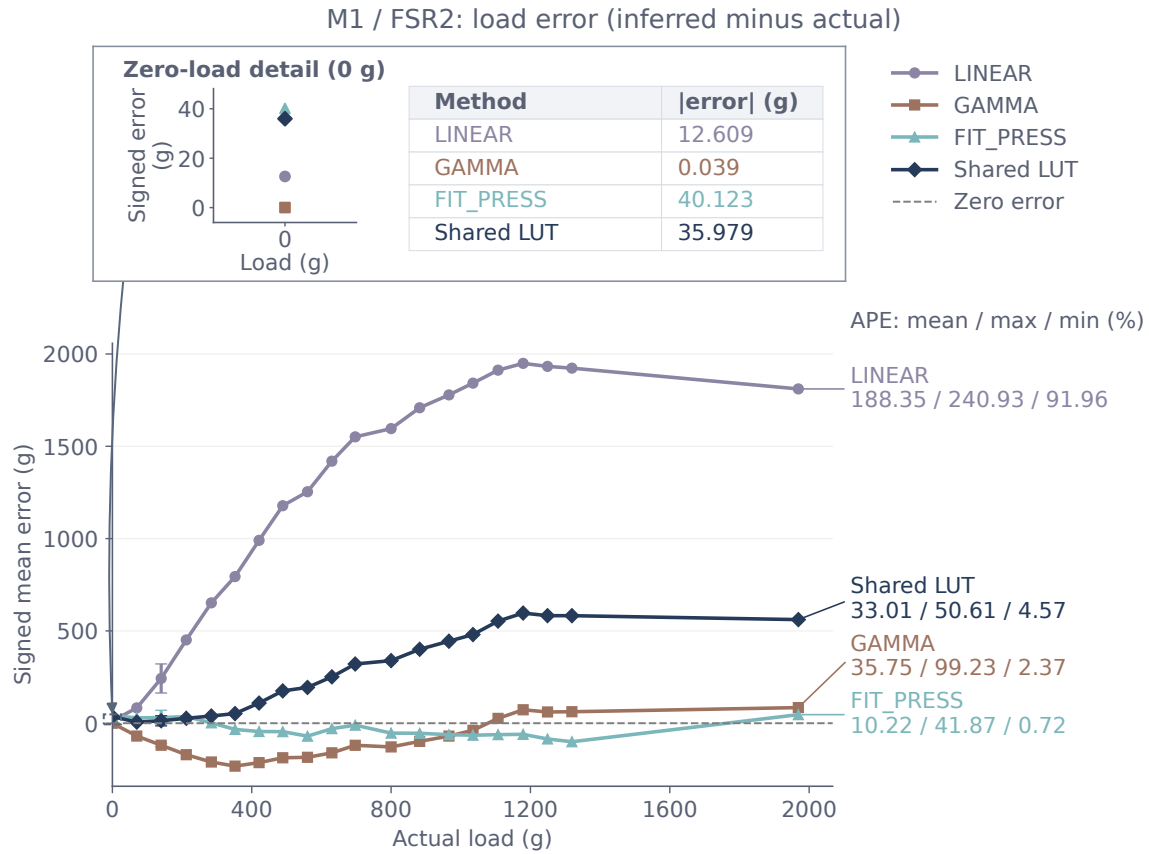


Figure 4.5: Signed error of the reconstructed layer mean. The 140 g point shows sample SD for $n = 2$; all other loads have $n = 1$. Percentage error is undefined at zero.

4.3.3 M1/FSR2 spatial dispersion

Against LINEAR, FIT PRESS had lower CV, variance, IQR and MAD at 9, 20, 19 and 19 of the 20 loads. Against GAMMA, the respective counts were 20, 18, 14 and 11. Excluding zero, they were 8/19, 19/19, 19/19 and 19/19 against LINEAR, and 19/19, 18/19, 14/19 and 11/19 against GAMMA.

With 22 masses, the layer-wide procedure requires 44 matrix captures per two-layer module, compared with 11,264 cell-wise captures. FSR1 was calibrated separately and is reported in Appendix E.2.

M1 / FSR2: dispersion of calibrated load estimates

—●— LINEAR —■— GAMMA —▲— FIT_PRESS

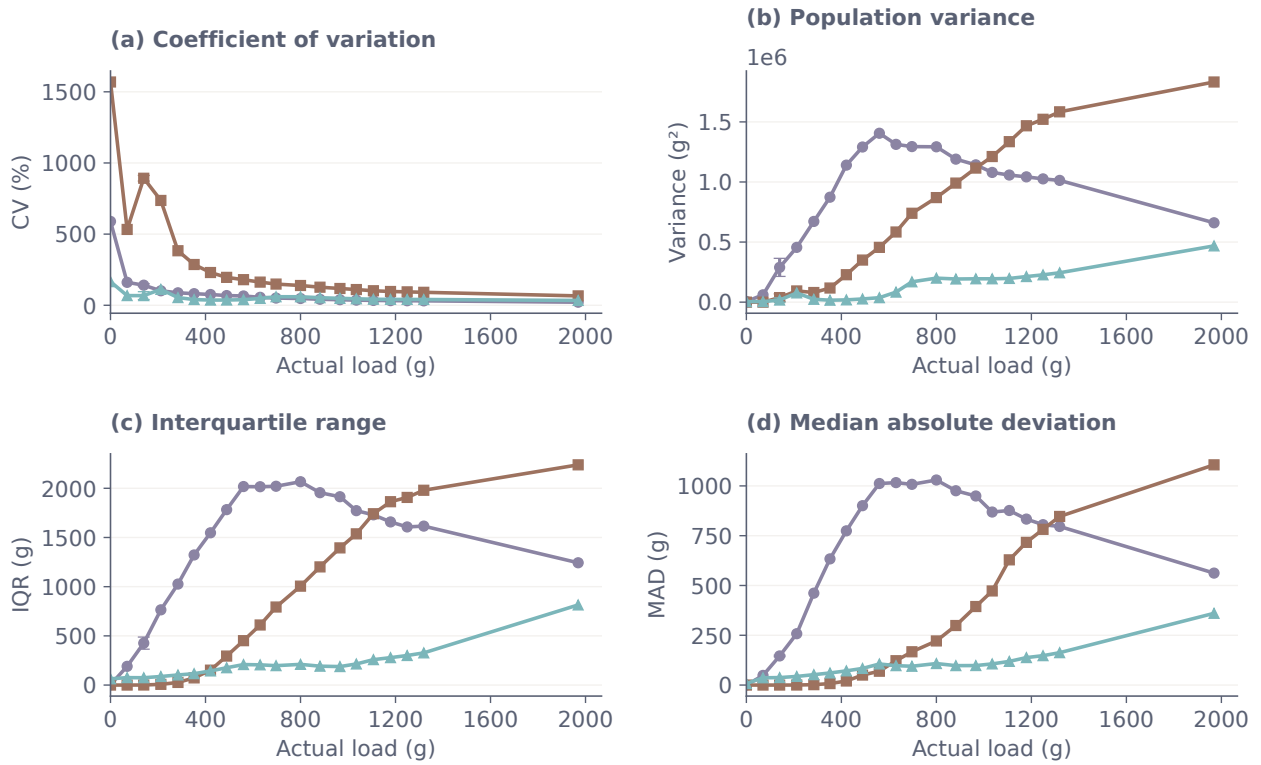


Figure 4.6: CV, population variance, IQR and MAD of the 256 cell estimates. Only 140 g has a repeat error bar ($n = 2$).

Chapter 5

Discussion

5.1 Data scalability

D-H1 is supported over the measured $N = 1\text{--}4$ domain. The observed *FULL* packet lengths matched the fixed 1044 B per-module frame plus the four-slot 40 B wrapper exactly. At the nominal 200 Hz reference, mean USB-rate errors were +0.029 to +0.125%; the small positive differences follow the measured mean round rates of 200.059–200.251 Hz. The $N > 4$ line uses $M = \max(4, N)$ and is a protocol-model extrapolation requiring additional addressing and scheduling, rather than validation of a 100-module system. The capacity intersections for the *FULL*, zero-load- K and rolling- K lines are reported in Figure 4.1 and the machine-readable audit in Appendix C.

The *DELTA* ledger includes ordinary mask population, observed module coverage and every recorded *ESKF*; its zero residual verifies representation consistency rather than an independent prediction of activity. Ordinary zero-load K increased from 8.26 to 10.04 per active module between $N = 1$ and $N = 4$. Module identity and module count are not separated by the available combinations, so a constant- K extension remains conditional. This interpretation is bounded by Table 4.1, Figure 4.1 and the per-run records in Appendix C.

D-H2 is supported as a bandwidth result for the five dynamic batches included in the performance analysis (Table 4.2; Appendix C).

The three windows per condition quantify short-term temporal variation within one setup. They do not estimate variation between assemblies, and the same four modules recur across the slot combinations. No throughput measurements beyond four modules, longer runs, or independent reassembly tests are included in the reported dataset (Table C.3; Table C.4; Figure C.1).

5.2 Power scalability

P-H1 receives descriptive support within the compiled $N = 1\text{--}4$ matrix. The multi-module currents differ from sums of the four separately measured module references by at most 1.483% under ZERO and 2.726% under MAX. The comparison is close to an additive construction: all eleven predictions reuse the same $N = 1$ references, and the $N = 2$ and $N = 3$ groups contain different module subsets. Each exact configuration was assembled once, so the residuals do not establish repeatability or a universal current-versus-count law. The values and residual range are shown in Table 4.3 and Figure 4.2; individual prediction entries are retained in Appendix D.

P-H2 is supported only for the recorded static readings. The lowest accepted DMM minimum remained 101 mV above the project lower limit, while the retained 3.224 and 3.232 V contact-fault readings show sensi-

tivity to connector condition. Sequential DMM readings cannot resolve simultaneous voltage sag, startup, ripple or load-switching transients. The MAX label also lacks a verified common mechanical load and is not a design worst case (Figure 4.3; Appendix D).

The reported power is the regulator output estimate for the module branches; it excludes regulator loss, the Teensy and the PC. No load sweep, controlled ambient record or thermal-equilibrium test was performed. The data therefore characterise measured operating points but do not determine transient supply capacity (Table 4.3; Figure 4.3; Appendix D).

5.3 Calibration scalability

For C-H1, FIT PRESS has lower FSR2 variance, IQR and MAD than LINEAR at nearly every non-zero validation load, but lower CV at only 8 of 19 loads. Against GAMMA, the corresponding counts are 18, 14, 11 and 20. The measured comparison therefore supports a metric-dependent rather than uniform reduction in spatial dispersion (Table 4.4; Figure 4.6).

For C-H2, FIT PRESS reconstructs the FSR2 layer mean more closely than the other three methods over the measured range, but averaging hides cell-level errors. Its 10.22% mean APE and 47.24 g mean absolute layer bias coexist with a 171.69 g mean per-cell MAE. At 1969 g, the layer-mean bias is +46.43 g while spatial SD is 684.23 g. These quantities quantify equivalent whole-layer mass reconstruction; local cell force and pressure were not measured (Table 4.4; Figure 4.5).

The 40.12 g FSR2 zero-load estimate is a systematic lower-end offset, and percentage error is undefined at zero. Ninety cell LUTs have a plateau touching zero, with a mean lower-end inverse value of 19.67 g, so midpoint assignment contributes a positive reconstruction floor. Across the layer, 170 flat intervals have a median width of 73 g and a maximum of 3855 g. Loads sharing one ADC knot cannot be distinguished by the inverse LUT, and separate validation ends at 1969 g despite the 5000 g training endpoint (Figure 4.4; Figure 4.5; Figure 4.6).

FSR1 demonstrates that the procedure can be applied to the second layer of the same module. Its Shared LUT has slightly lower layer-mean APE than FIT PRESS (15.09 versus 15.57%), whereas FIT PRESS has lower mean per-cell MAE (174.98 versus 382.47 g). This second application does not test transfer of a model between layers, another module or a reassembled sensor. With one capture at nearly every load, between-session precision remains unmeasured (Appendix E.2; Table E.2; Figure E.3).

5.4 Ethical considerations

The reported experiments were benchtop tests of FSR hardware and communication records. The retained acquisition inventory contains no human-participant, animal or personal-data records (Appendix C).

Raw recordings, diagnostic batches and source-path manifests remain preserved, while the main results distinguish selected performance windows from CHECK records (Appendix C). The projected module limits in Figure 4.1 are labelled as conditional reference-ceiling calculations rather than measured safety or reliability limits.

The calibration output is an equivalent whole-layer mass estimate, not a validated local pressure measurement. The measured error and spatial spread are reported in Figure 4.5 and Figure 4.6; transient power behaviour is not measured (Appendix D).

The current records contain slot labels and experiment metadata but no personal identifiers. This report there-

fore contains no participant-data or deployment privacy analysis beyond that record-level scope (Appendix C).

No lifecycle, electronic-waste or environmental-impact measurements were made in this report; those quantities are outside the communication, power and calibration records (Appendix A).

5.5 System-level interpretation

The prototype provides measured scaling evidence over one to four modules. *DELTA* reduces transported bytes but retains activity-dependent demand and a separate delivery-integrity requirement. Power measurements describe approximately additive static demand without establishing transient reserve. Layer-wide calibration reduces acquisition count and was applied to both FSR layers of M1, while its accuracy and dispersion remain layer- and metric-dependent (Figure 4.1; Table 4.3; Table 4.4; Figure 4.6).

The extrapolations are planning models rather than additional experiments. The reported records contain no sustained SPI timing or cache-recovery test, synchronised transient-power measurement, cross-module calibration, or reassembly repeat. Interaction between the two FSR layers and the ACC sensor is also outside the present experiments (Figure 4.1; Appendix C; Appendix D; Appendix E.2).

Chapter 6

Conclusion

This report quantified data, static power and calibration measurements in a four-module FSR prototype. The experiment covered all 15 module combinations in *FULL* and *DELTA*, with three zero-load windows per combination. Four-module *FULL* traffic was 6.747828 ± 0.001148 Mbit/s. Measured *FULL* packet lengths matched the protocol model at $N = 1-4$, and mean USB rates differed from nominal 200 Hz predictions by +0.029 to +0.125%. The same model gives 167.7184 Mbit/s at $N = 100$; this value is an extrapolation beyond the measured range. Under the same 200 Hz conditions, using the 480 Mbit/s USB and 10 Mbit/s raw HOST SPI references, the theoretical intersections were $N = 286.24/5.99$ for *FULL*, $N = 464.25/41.22$ for the zero-load $K_0(N) = 7.554189 + 0.582684N$ projection, and $N = 501.41/10.52$ for a rolling stress line fixed at the observed ordinary maximum $K = 254$. The corresponding largest integer counts were 286/5, 464/41 and 501/10 (USB/HOST SPI), as shown in Figure 4.1; they are conditional planning values. *DELTA* reduced same-rate packet bytes by 86.60–88.73% at zero load and 65.10–80.57% in five passing dynamic batches. The 30 compiled power conditions gave four-module currents of 72.513 mA for ZERO and 78.720 mA for the recorded MAX label; the lowest accepted DMM minimum was 3.236 V. These static, sequential measurements do not establish transient supply capacity. Separate 22-point sweeps calibrated both FSR layers of M1. FSR2 FIT PRESS gave 10.22% mean layer-estimate APE and 171.69 g mean per-cell MAE over 21 validation captures, with a 40.12 g zero-load estimate and metric-dependent dispersion. FSR1 confirmed application of the procedure to a second layer but not model transfer or cross-module repeatability. The reported dataset contains no measurements beyond four modules, no transient power measurement, and no cross-module or reassembly calibration repeat.

Chapter A

Current hardware design records

A.1 Hardware renderings

The following composite figure combines the latest hardware renderings supplied in docs/Final/hardware/. It documents the physical design described in Section 3.1. The accelerometer panel records the existing prototype/interface only; it is outside the scope of the present study and is not used in any reported evaluation.

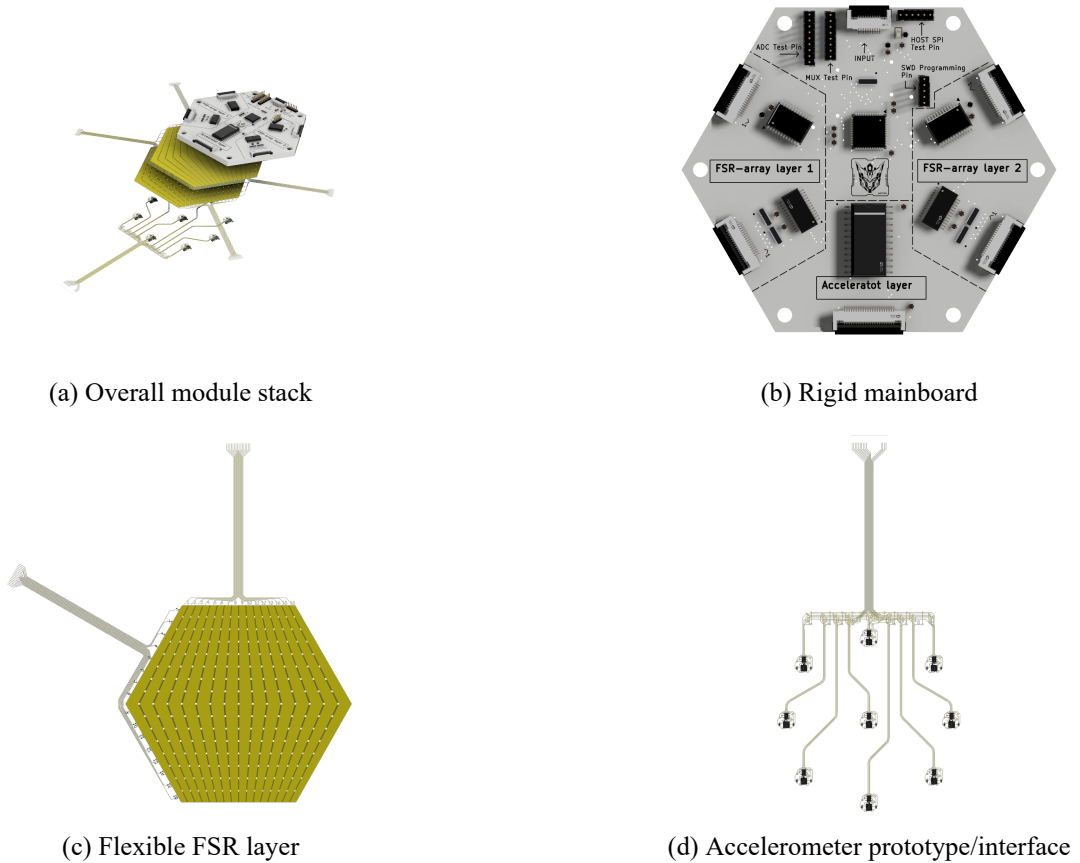


Figure A.1: Current hardware renderings.

A.2 Schematics and PCB layouts

The following figures reproduce the current exported schematic and PCB-layout records for the rigid mainboard, flexible FSR layer and flexible ACC layer. They document the implemented design files; inclusion here does not add an electrical-performance claim. The ACC records describe the existing hardware prototype only and are not used in the reported experiments.

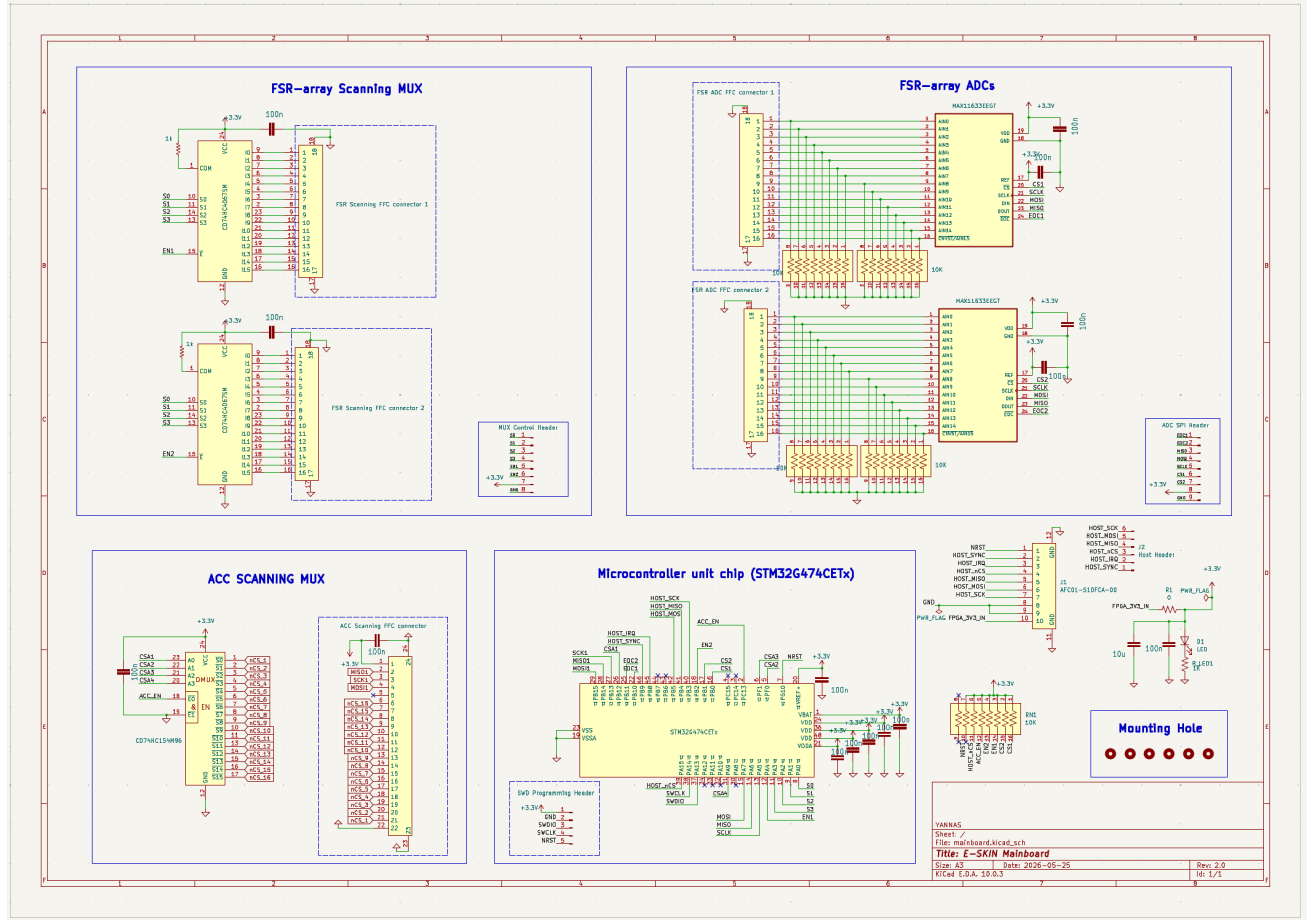


Figure A.2: Current rigid-mainboard schematic, including the two FSR scanning paths, ADC interfaces, ACC interface, STM32 controller and host connection.

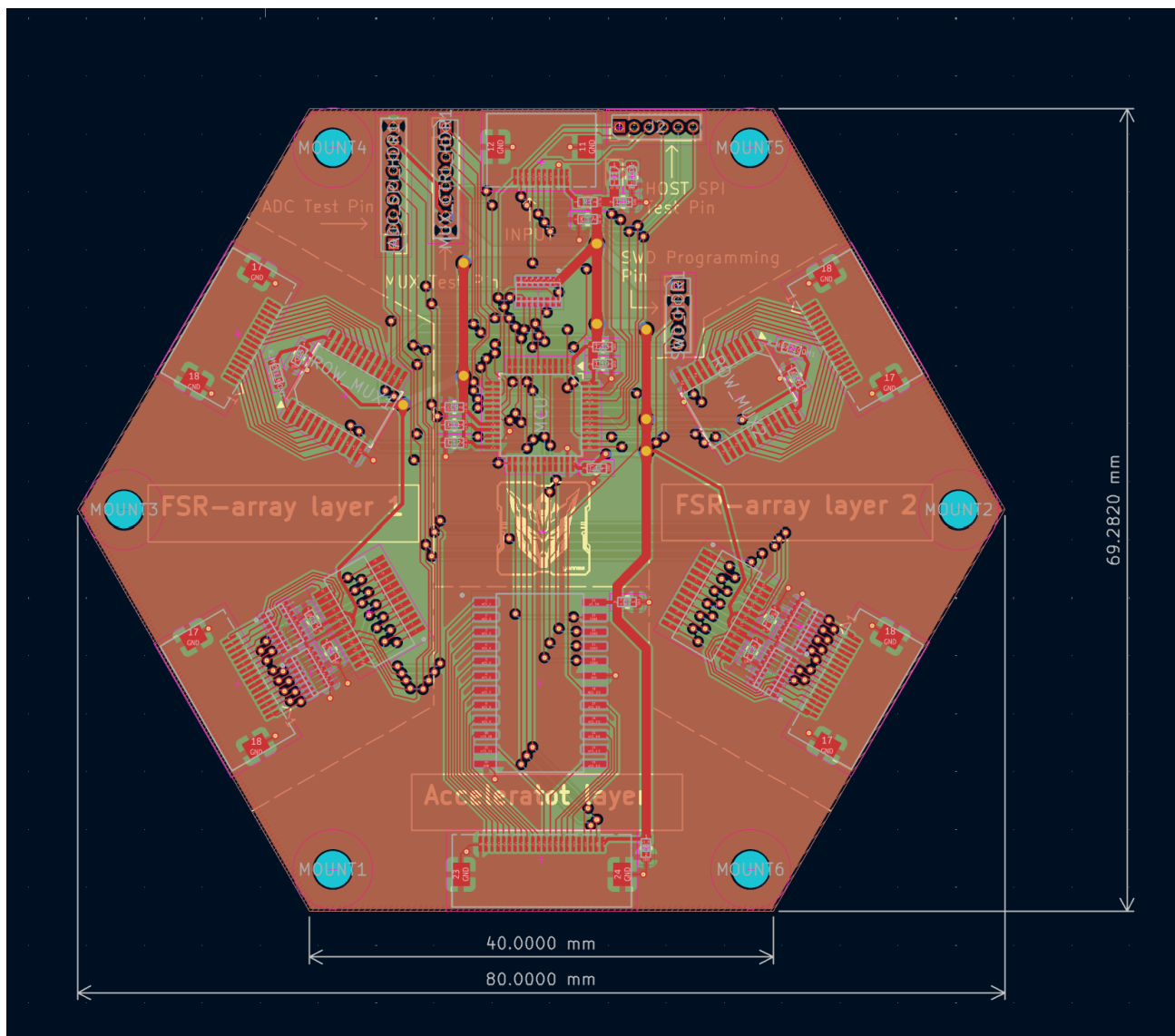


Figure A.3: Current four-layer rigid-mainboard PCB layout, showing the hexagonal outline, component placement, routing and mounting geometry.

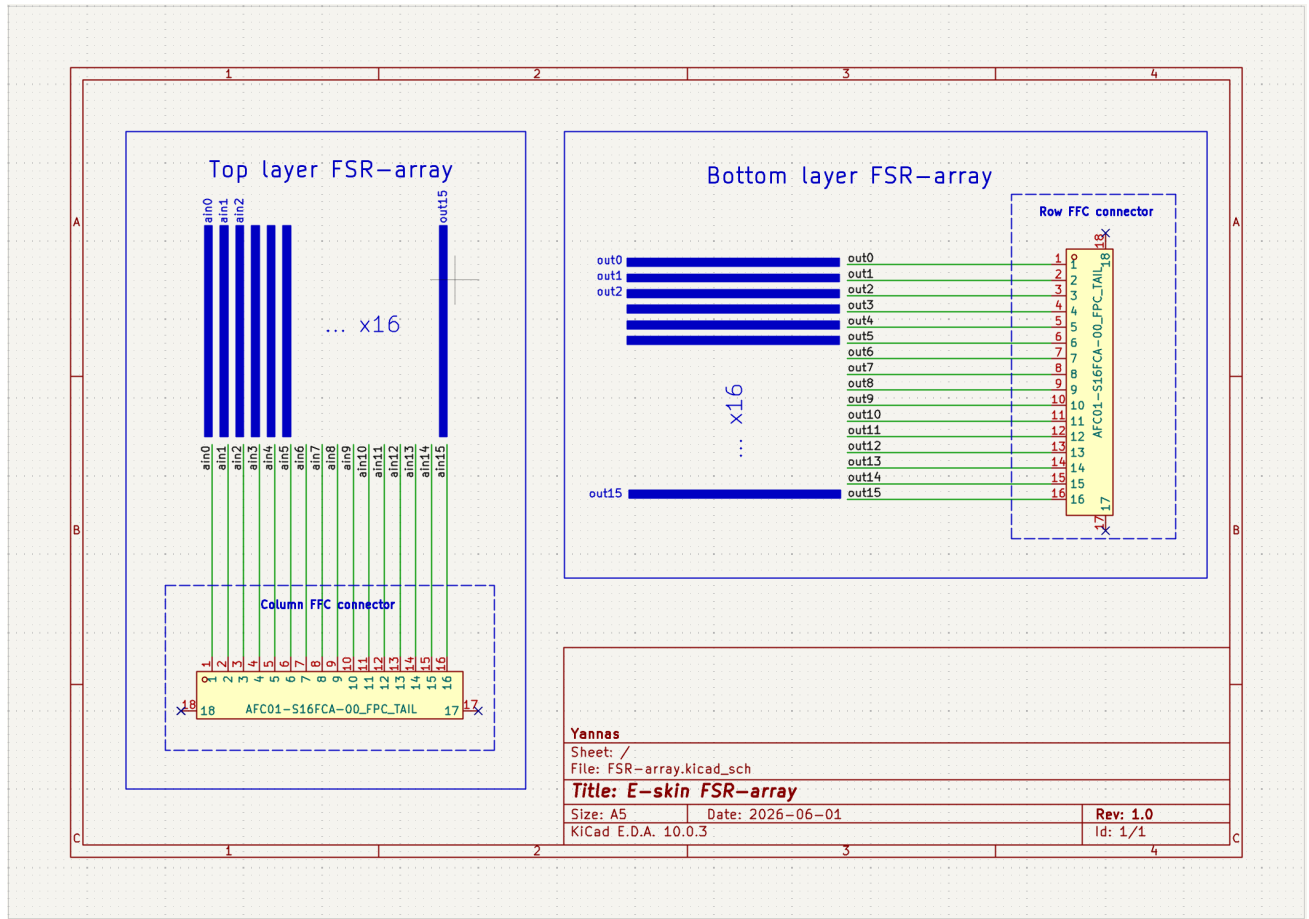


Figure A.4: Current flexible FSR-layer schematic. The orthogonal sets of 16 electrodes form the 16×16 sensing matrix and connect through separate row and column FFC tails.

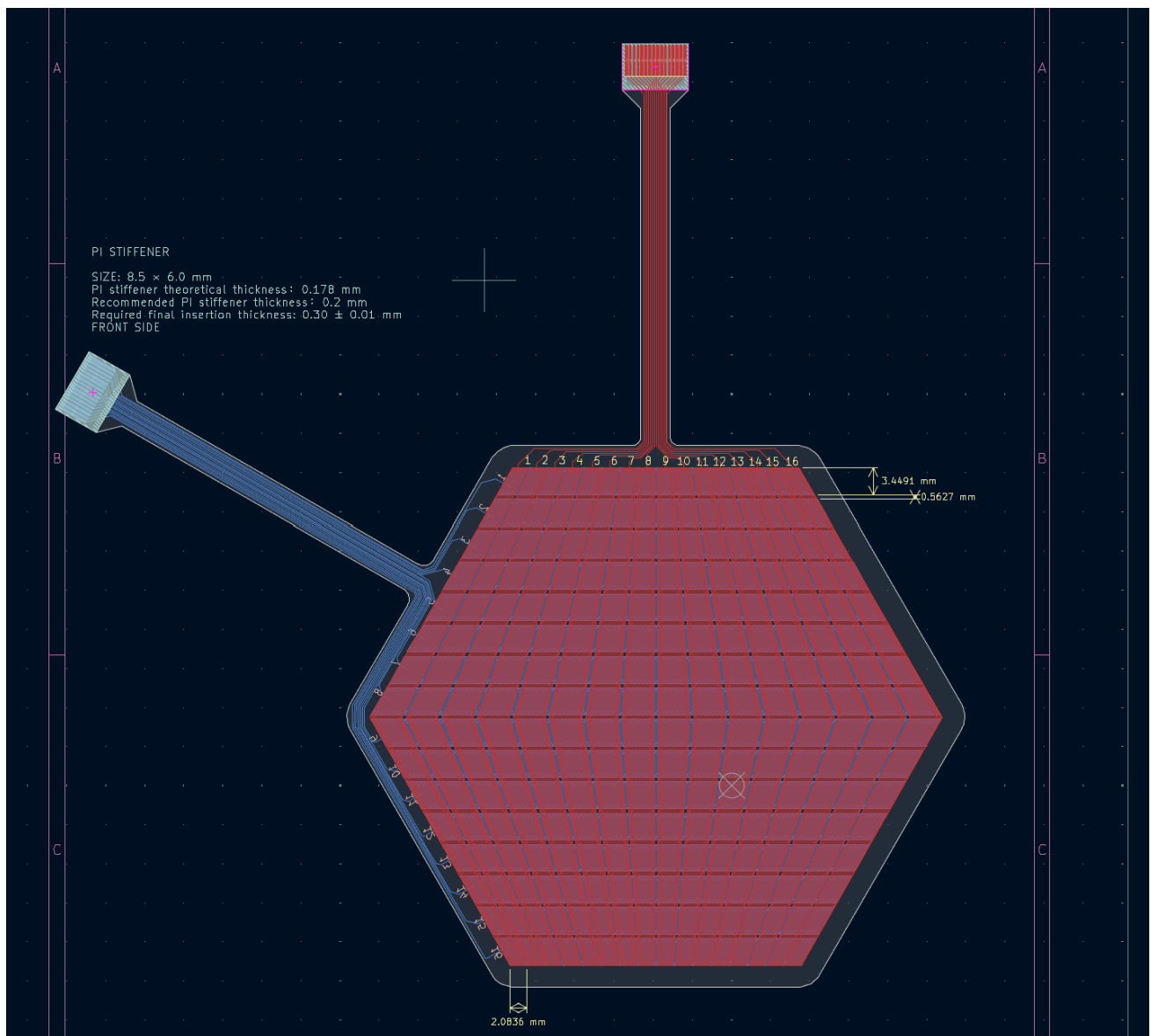


Figure A.5: Current two-layer flexible FSR PCB layout, including the matrix electrodes, hexagonal outline and two FFC tails.

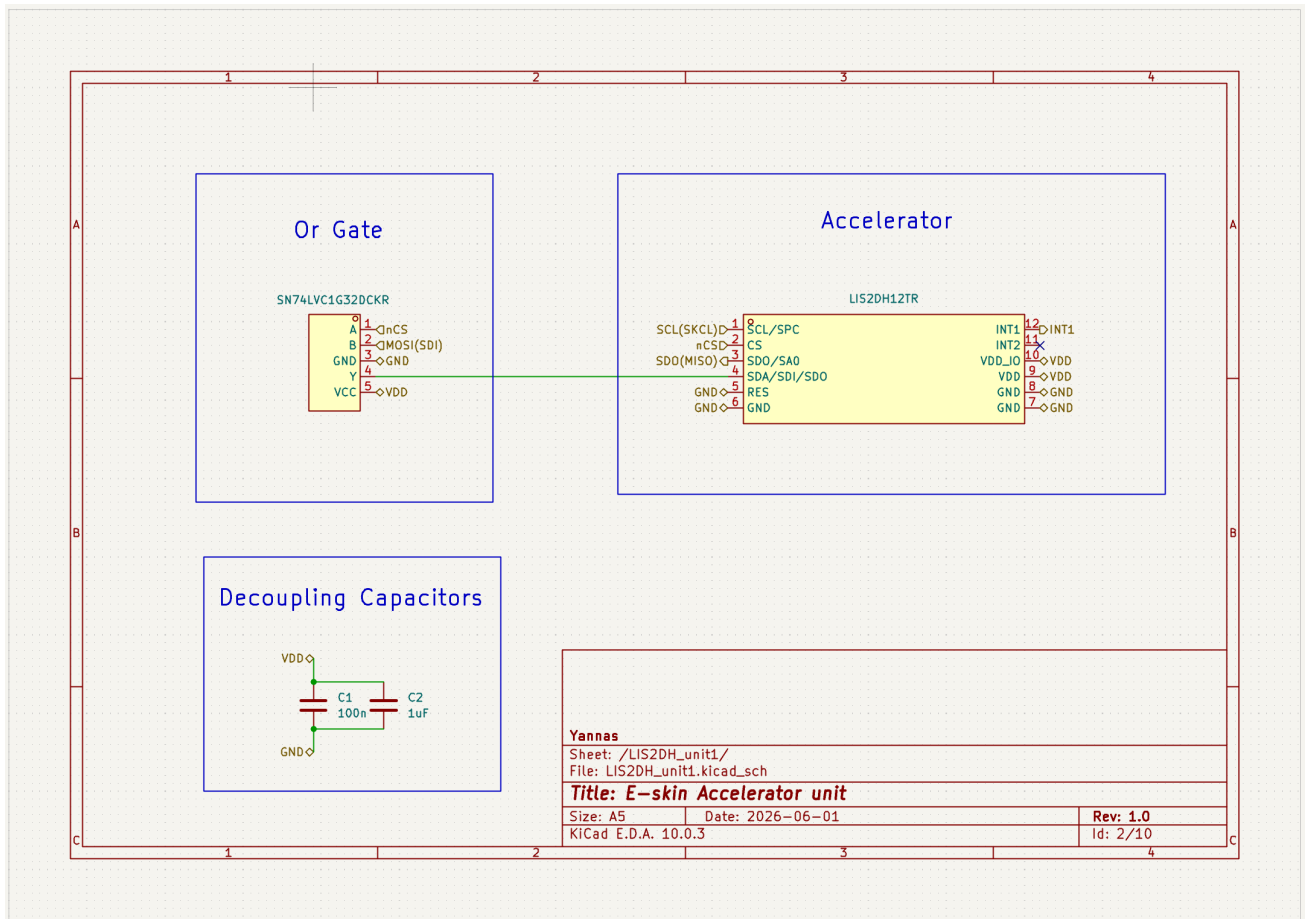


Figure A.6: Current flexible ACC-layer schematic, showing the accelerometer, logic interface and local decoupling.

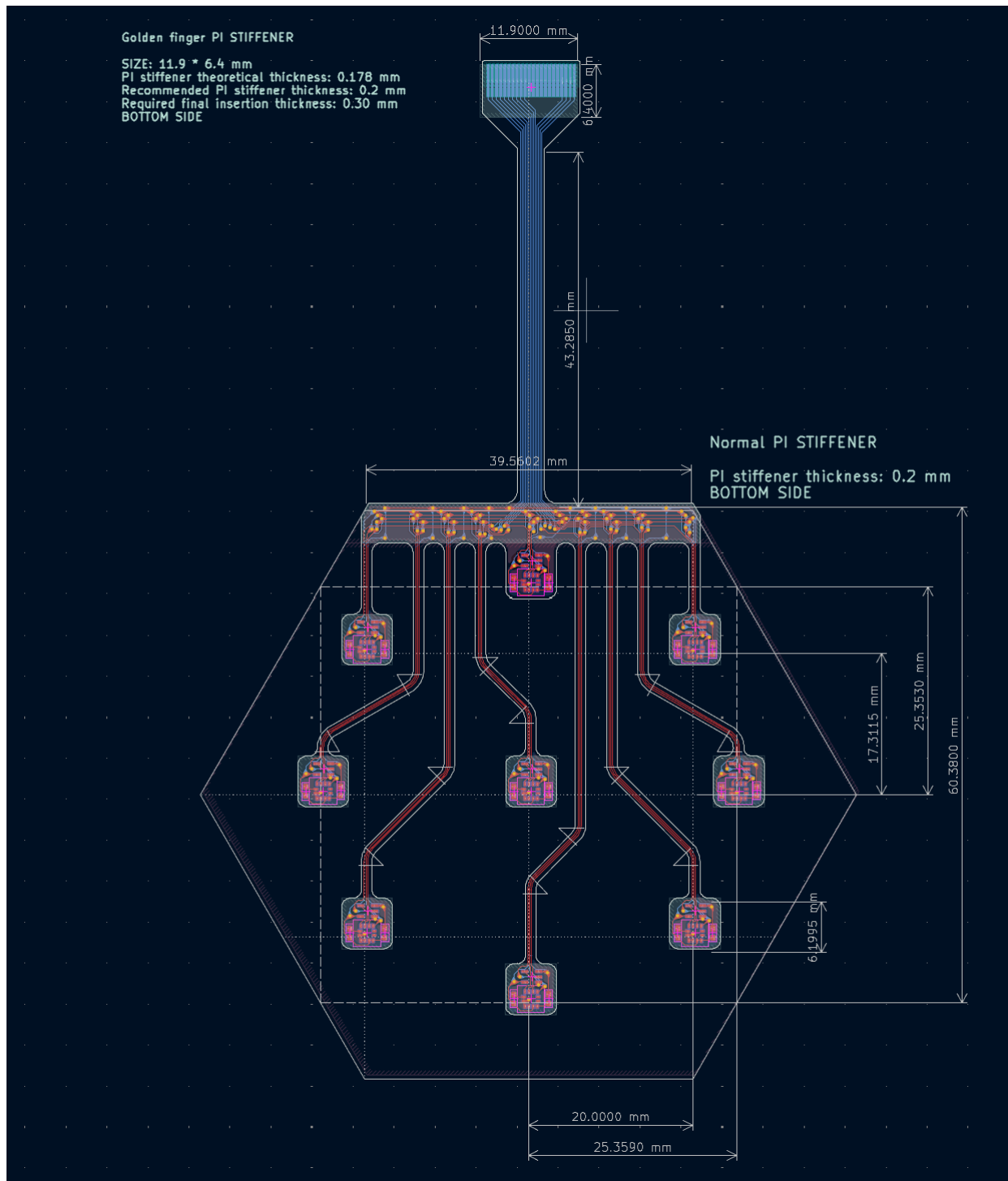


Figure A.7: Current flexible ACC-layer PCB layout, including the eight sensor positions, routing, board outline and FFC tail.

Chapter B

Complete protocol definitions

The current FSR-only design uses ESK version 4: 1044-byte full frames and mask-based updates. The earlier position-record DELTA layout is legacy and is not used in the current throughput model. Numeric fields are little-endian, byte offsets start at zero, and CRC32 is IEEE CRC32 over all preceding bytes in the frame. The current variable-length implementation clocks the complete encoded *ESKF* or *ESKD* frame for the selected module, including the 16-byte ESK header and inner CRC. No unused buffer-tail bytes are clocked. USB then carries these frames inside its own *MULI* v2 wrapper. The 6–7 September 2026 campaign is associated with this revised firmware; its local configuration records give a 10 MHz Host SPI setting, consistent with the operator’s reported successful tests. The stored firmware hashes identify local build artefacts, not firmware read back from each device. USB captures verify delivered packet contents and lengths, while SPI transaction timing and unsuccessful reads require separate line traces. The 40 MHz reference used for capacity calculations remains untested.

B.1 Protocol stack

Table B.1: Protocol layers.

Layer	Direction	Content
Host command	Teensy → STM32	16-byte <i>DSCM</i> command carried in the MOSI bytes of the next Host SPI read.
Module frame	STM32 → Teensy	One logical <i>ESKF</i> or mask-based <i>ESKD</i> frame.
Module block	Teensy internal assembly	4-byte module metadata followed by the encoded module frame or an error-diagnostic prefix; an empty slot has zero payload length.
<i>MULI</i> / USB	Teensy → PC	20-byte <i>MULI</i> header, M module slots with their actual payload lengths, and a 4-byte outer <i>CRC32</i> ; currently $M = 4$.

The implemented Host SPI sequence is: the STM32 encodes a frame of length L , arms one transmit/receive DMA for L bytes and raises its ready IRQ. The Teensy asserts that module’s chip select and exchanges 16 bytes: the *DSCM* command on MOSI and the ESK header on MISO. After checking the header, it clocks $L - 16$ more bytes while chip select remains low, then releases chip select. The complete transaction clocks L bytes; neither the header nor the simultaneous command is counted twice. An invalid header ends the read after 16 bytes and the STM32 recovers the interrupted DMA.

B.2 Host command (*DSCM*)

Table B.2: Host SPI command.

Offset	Field	Meaning
0–3	<i>DSCM</i>	Command marker.
4	Version	Command version 2.
5	Mode	0 = <i>FULL</i> , 1 = <i>DELTA</i> .
6–7	Delta threshold	ADC-code threshold; zero keeps the current setting, initially 8.
8–9	Reserved	Zero-filled; field offsets are unchanged.
10–11	Scan rate	Requested scan frequency in Hz.
12	Command flags	Bit 0 requests <i>DELTA</i> re-synchronisation.
13–15	Reserved	Zero-filled.

The command is not a separate transaction. It occupies the first 16 MOSI bytes of the next read while the STM32 returns the previous or newly prepared module frame on MISO.

B.3 *ESK* frames: full synchronisation and *DELTA*

Current *ESK* frames use the common 16-byte header below.

Table B.3: *ESK* header.

Offset	Field	Meaning
0–3	Marker	<i>ESKF</i> or <i>ESKD</i> .
4	Version	Protocol version 4.
5	Flags	Payload and integrity flags.
6–7	Length	Total logical frame length in bytes, including the trailer.
8–11	Sequence	Module frame counter; continuity is checked across both frame types, allowing counter wrap but reporting resets or jumps.
12–15	Base sequence	Sequence of the cache used by a <i>DELTA</i> update; 0xFFFFFFFF for a full synchronisation.

Table B.4: *ESK* frame layouts.

Frame	Role	Payload after the 16-byte header	Length
<i>ESKF</i>	Full synchronisation	FSR1: 512 B; FSR2: 512 B; inner <i>CRC32</i> : 4 B.	1044 B
<i>ESKD</i>	Mask-based update	FSR1 mask: 32 B; FSR2 mask: 32 B; <i>K</i> updated values of 2 B each; inner <i>CRC32</i> : 4 B.	$84 + 2K$ B
<i>ESKD</i> , $K = 0$	No-change update	Two zero masks: 64 B; inner <i>CRC32</i> : 4 B.	84 B

There is no separate change-count field: K is the number of set bits across the two masks. Values follow those bits in FSR1-then-FSR2, row-major order.

Table B.5: Mask-based *DELTA* payload.

Offset	Field	Meaning
16–47	FSR1 mask	256 bits, one per taxel.
48–79	FSR2 mask	256 bits, one per taxel.
80 onward	Values	K little-endian 16-bit ADC values.
$80 + 2K$	Trailer	4-byte <i>CRC32</i> .

For zero-based row r and column c , position $p = 16r + c$ uses bit $p \bmod 8$ of mask byte $\lfloor p/8 \rfloor$, least-significant bit first. The PC accepts an *ESKF* base and applies the mask-selected values to its cache. A base mismatch prevents further updates until full re-synchronisation. In *DELTA* mode, the encoder checks at scan encoding whether at least 1000 ms have elapsed since the last *ESKF*. If so, that scan is encoded as a periodic complete frame. Initialisation, a mode change that invalidates the baseline, an explicit re-synchronisation request, and the $K > 480$ fallback can also produce *ESKF*. Every *ESKF* resets the timer, regardless of its cause. It replaces that scan’s ordinary update; it is not an extra packet added between scans. The 1000 ms condition is checked when encoding occurs, so it does not guarantee an independently scheduled transmission at an exact wall-clock interval.

At $K = 480$, an ordinary *ESKD* and a complete *ESKF* both occupy 1044 B. An *ESKF* carries no change mask or trigger code. Its marker therefore identifies a complete frame but does not establish that 480 taxels changed, or distinguish periodic synchronisation from initialisation, requested recovery or high activity. The recorded total complete-frame fraction can be measured; its trigger categories remain unknown without correlated firmware events, as specified in Equation 3.8.

The independent audit keeps two separate states: the most recently observed module sequence, and the cache sequence used to apply the next delta. An *ESKF* supplies a new valid cache base, but does not erase a discontinuity relative to the preceding observed inner sequence. A following *ESKD* must reference the current cache base and pass mask, length and CRC checks before it is applied. Initial deltas observed before any full anchor in a recording are reported separately from deltas that cannot be applied after an established anchor. An *ESKF* restores the current state; it does not retransmit intervening missing measurements.

ESK0 and the earlier 4-byte position records are legacy formats, not current mask-based packets. The current integrity and *DELTA* flags are 0x10 and 0x20, respectively.

B.4 *MUL1* and USB module blocks

The Teensy wraps the encoded module frames for USB delivery using *MUL1* version 2. The packet retains M slot descriptors, where M is the configured maximum module count and is four in the current project. Each slot’s 2-byte payload-length field locates the next descriptor; the reserved field in the outer header remains zero.

Table B.6: *MULI* header.

Offset	Field	Meaning
0–3	Marker	<i>MULI</i> .
4	Version	<i>MULI</i> version 2.
5	Slot count	Configured module positions M , currently 4.
6	Updated mask	Module-update bitmap.
7	Reserved	Reserved.
8–9	Declared length	Total <i>MULI</i> length in bytes.
10–11	Reserved	Zero; each slot descriptor is defined separately as 4 B.
12–15	Sequence	Host packet sequence.
16–19	Host timestamp	Teensy bridge timestamp in milliseconds; distinct from the PC packet-completion timestamp.

Table B.7: *MULI* module block.

Offset	Field	Meaning
0	module_id	Zero-based CS/IRQ slot index (0–3), not a physical-board unique identifier.
1	Status	Zero for an updated valid frame; non-updated slots use bit 7 plus a read-result code.
2–3	Payload length	Number of valid bytes following this 4-byte metadata.
4 onward	Slot payload	Complete <i>ESKF/ESKD</i> frame, or diagnostic prefix for selected errors; no padding.

The total USB packet length is therefore

$$L_{MULI} = 20 + \sum_{i=1}^M (4 + L_i) + 4 = 24 + 4M + \sum_{i=1}^M L_i \text{ B}, \quad (\text{B.1})$$

where L_i is the actual payload length carried by slot i . An empty, non-updated slot has $L_i = 0$, while selected errors carry a 16 B diagnostic prefix. The final 4 B are the outer *CRC32*. Thus the outer overhead is $24 + 4M$ B for both modes, with $M = 4$ giving 40 B regardless of the number of active modules N . For a successful ordinary-update round with K changes in each of $N \leq M$ modules, the actual Host SPI frame bytes total $N(84 + 2K)$ B, while USB carries $24 + 4M + N(84 + 2K)$ B. A complete-frame round instead uses $1044N$ B on SPI and $24 + 4M + 1044N$ B on USB. Mixed frame types are counted using their actual lengths.

With $M = 4$, a complete *FULL* round has length 1084, 2128, 3172 or 4216 B for $N = 1, 2, 3, 4$. An all-*ESKD* round with $K = 0$ in every active module instead has length 124, 208, 292 or 376 B. These are different from a 40 B outer packet containing four empty slot descriptors: that packet can have a valid outer *CRC* while providing no new measurement from any target module. A zero change mask is a valid no-change update; a missing slot payload is an absent update.

For the measured $N = 2$ case, the two slots can carry different frame types. A D/D packet has length $208 + 2(K_1 + K_2)$ B; a D/F or F/D packet has length $1168 + 2K_D$ B; an F/F packet has length 2128 B. Thus a packet containing one *ESKF* must not automatically be counted as an all-*FULL* packet. The mean model weights each slot’s ordinary-frame mean K , total *ESKF* fraction and observed update coverage separately, and the finite byte ledger in Equation 3.10 counts every received payload, including retained diagnostic prefixes.

Outer packet sequence gaps, module sequence jumps and absent target-slot updates are reported separately.

A valid CRC tests the bytes in a received frame, not the existence of frames that were not observed. Likewise, successful reconstruction after a synchronisation frame does not establish uninterrupted delivery before it. Recorded PC and bridge timing can help locate the observed interruption, but cannot alone determine the number of physical scans lost or identify SCK, a high- K fallback or a periodic *ESKF* as its cause.

Larger logical-round projections require $M \geq N$ and protocol/host extensions beyond the current four-slot implementation. The current one-byte update bitmap and two-byte length field do not by themselves implement the module counts shown in large-scale theoretical plots; additional addressing and fragmentation overhead is outside those plots.

Chapter C

Data scalability records

The current measurements come from the 6–7 September 2026 revised-firmware campaign in `docs/Final/data/scalability/data/`. `run_index.csv` identifies all 114 records, their original and corrected conditions, slot combinations, repeat and batch identifiers, current directories and integrity findings. `primary_matrix.csv` selects 90 zero-load records; `dynamic_analysis_selected.csv` selects 15 dynamic records from five complete passing batches. `analysis_selected.csv` contains their union of 105 records. This is an index of eligible evidence, not a population to pool across conditions.

The complete report source, acquisition records, analysis scripts, generated figures and audit outputs are archived in the project repository: https://github.com/Yannas-Sun/E-skin_MSc_final_report. Paths in this appendix are relative to that repository release.

Each record retains six source files: `mul1_raw.bin`, `packet_log.csv`, `module_log.csv`, `events.jsonl`, `summary.json` and `experiment_log.md`. Run identifiers include a date, time and fractional-second suffix; the original `run_id` and `batch_id` remain authoritative after classification. The 684 source files were hash-checked during final classification. `path_map.csv` resolves source paths, including acquisition-time paths embedded in unchanged logs. Figures and independent audit outputs are derived artefacts, not replacements for these sources.

The final evidence snapshot is the `20260907_final_data_review/` folder in the `Data analysis/` directory, in particular `evidence_review.json` and `FINAL_SUMMARY_CN.md`. Per-batch raw audits and per-slot statistics are in the neighbouring analysis directories. The 1 and 5 September records remain in the `Data history/` directory, within `pre_variable_spi_20260906/data/`. They provide historical provenance and are not pooled into current measurements or error bars. The post-reinstallation four-module review is retained in `20260907_n4_connector_retest_review/`; its fixed-run raw audit, condition correction, organization plan and hash verification distinguish the new performance batch from the preserved pre-reinstallation diagnostic.

C.1 Measured and derived variables

All main-window quantities use the same complete, valid *MUL1* packets completed from 10 s to recording stop. Each capture lasts approximately 40 s, leaving approximately 30 s for the main analysis. The retained first 10 s and partial boundary bytes are assessed separately.

Table C.1: Variables used for the revised-firmware data analysis.

Variable	Source, definition and use
N , target slots, M	Configuration and audit: expected active slots; $M = 4$ fixes outer overhead at $24 + 4M = 40$ B for every N .
f_{target} , Δ	Recorded settings: 200 Hz commanded rate and a <i>DELTA</i> threshold of 8 ADC codes.
T	PC recording clock: main duration from 10 s to stop, not a selection of error-free intervals.
P , B	Parsed complete valid outer-packet count and summed declared lengths in the same window. Raw arrival bytes and boundary fragments are retained separately.
f_{eff} , $\bar{L}$, $R_{\text{USB, meas}}$	Derived P/T Hz, B/P B and $8B/(10^6 T)$ Mbit/s, respectively; see Equation 3.11.
$K_{D,i}$	Set bits across both valid <i>ESKD</i> masks for slot i ; <i>ESKF</i> supplies no measured K .
q_i , c_i	Complete-frame fraction and observed update coverage, calculated from valid inner-frame counts with distinct denominators.
L_i , B_{other}	Actual slot lengths and bytes outside valid ESK frames, including retained diagnostics.
Integrity findings	CRC/layout validity, outer and per-slot sequence continuity, target updates and cache chain; inner sequences include transitions through <i>ESKF</i> .
PC/bridge intervals	PC completion and bridge packet clocks; neither directly measures SPI clocking or physical scans.
f_{SCK}	Operator-tested 10 MHz and local configuration provenance; 40 MHz remains theoretical. No device readback or waveform is supplied by these records.
SPI frame bytes	Protocol-derived sum of received ESK lengths; excludes the USB wrapper, unobserved transactions and retries.
Reduction	Complete-packet byte saving against <i>FULL</i> at the same observed packet rate, Equation 3.12.

C.1.1 Complete-frame fractions, coverage and averaging

For slot i , $q_i = n_{F,i}/(n_{F,i} + n_{D,i})$ uses valid inner frames in the window. In contrast, $c_i = (n_{F,i} + n_{D,i})/P$ uses observed outer packets and measures whether that slot updated. A pooled complete-frame fraction is $\sum_i n_{F,i} / \sum_i (n_{F,i} + n_{D,i})$. The fraction of outer packets containing at least one *ESKF* has denominator P and is reported separately. One slot can carry *ESKF* while another carries *ESKD*; these fractions are not interchangeable with the fraction of all-FULL packets.

The ordinary-update mean $\bar{K}_{D,i}$ uses only *ESKD* mask populations. It is undefined when $n_{D,i} = 0$; q_i is undefined when there are no valid inner frames. A slot with only *ESKF* has a mean valid-frame length of 1044 B. Empty slots and invalid or diagnostic payloads are not zero-change updates. Equations 3.9 and 3.10 include their received bytes without implying successful reconstruction. Missing outer packets remain a separate finding even when coverage within received packets is complete.

All *ESKF* frames contribute to measured byte cost. The encoder’s 1000 ms condition, initialisation, invalidated baseline, re-synchronisation request and high-activity fallback can lead to a complete frame. Each complete frame replaces an ordinary update and resets the timer. The wire format has no cause field: the observed total q_i cannot be partitioned exactly into periodic and exceptional causes. An interval near one second does not prove a periodic trigger. Per-slot bridge intervals and PC intervals remain distinct descriptive observations; merging event times across slots does not measure a module’s synchronisation period. The periodic-only $q = 0.005$ reference at 200 Hz is a model assumption.

Run-level means are calculated before aggregation. Reported uncertainty is the sample SD of window-level values with denominator $n - 1$. Each combination or dynamic batch has three consecutive windows within one setup. Frames are not experimental replicates, and independent setup repetitions are not established. Zero-load N -level means weight the combinations equally; with three windows per combination, this equals the across-window mean. The across-window SD contains both combination and temporal variation. At $N = 4$, only one slot combination exists, so no between-combination SD is available. A single-window diagnostic subset has no between-window SD; every selected dynamic batch contains three windows.

C.2 Experiment arrangement

Tables C.2–C.4 identify the configurations and selection. All records use the 200 Hz command and retain their complete capture. The fixed 10 s prefix is not a retrospective deletion of failure events; full-capture findings remain available beside the main-window assessment.

Table C.2: Complete zero-load matrix for each mode. Each subset has three consecutive windows.

N	Target slot subsets	Subsets	Windows/mode
1	M0; M1; M2; M3	4	12
2	M0+M1; M0+M2; M0+M3; M1+M2; M1+M3; M2+M3	6	18
3	M0+M1+M2; M0+M1+M3; M0+M2+M3; M1+M2+M3	4	12
4	M0+M1+M2+M3	1	3
Total	<i>FULL</i> and <i>DELTA</i> each cover all subsets	15	45

Table C.3: Inventory and selection by recorded window. PASS/CHECK refers to main-window findings, not independent trial probabilities.

Group	Retained	PASS	CHECK	Selected
<i>FULL</i> , zero load	48	47	1	45
<i>DELTA</i> , zero load	45	45	0	45
<i>DELTA</i> , dynamic	21	16	5	15
Total	114	108	6	105

The original *FULL* M0+M1+M2 batch has one CHECK and two PASS windows. All three remain available, but a complete replacement batch supplies that primary combination. For *DELTA*, the original GUI status is preserved alongside the independent review: cache acceptance at an *ESKF* does not establish inner sequence continuity. The 108 passing windows are not a reliability success proportion and are not the same selection as the 105 windows from complete eligible batches.

Table C.4: Dynamic batches on 7 September. Start time identifies the first capture of each three-window batch, using local recorded time.

Batch start	Slots	Loading scenario	PASS / CHECK	Analysis role
04:06:21	M1	Large area, initial batch	0 / 3	Diagnostic
05:04:53	M1	Large area, later batch	3 / 0	Performance
05:19:41	M1	Repeated press/release	3 / 0	Performance
05:38:58	M1	Rolling	3 / 0	Performance
05:47:52	M1+M3	Simultaneous large-area coverage	3 / 0	Performance
06:01:47	M0–M3	Simultaneous full coverage, before connector reinstallation	1 / 2	Diagnostic
11:51:45	M0–M3	Simultaneous full coverage, after connector reinstallation	3 / 0	Performance

Later dynamic labels were corrected using the operator’s confirmation. Original zero-load metadata are retained; force, contact area, exact load distribution and unspecified layer details are not invented. Slot labels identify CS/IRQ connections, not unique board serial numbers. These manual scenarios test observed activity without establishing matched mechanical loading between module counts. The current campaign has no separate small-area or $N = 3$ dynamic batch.

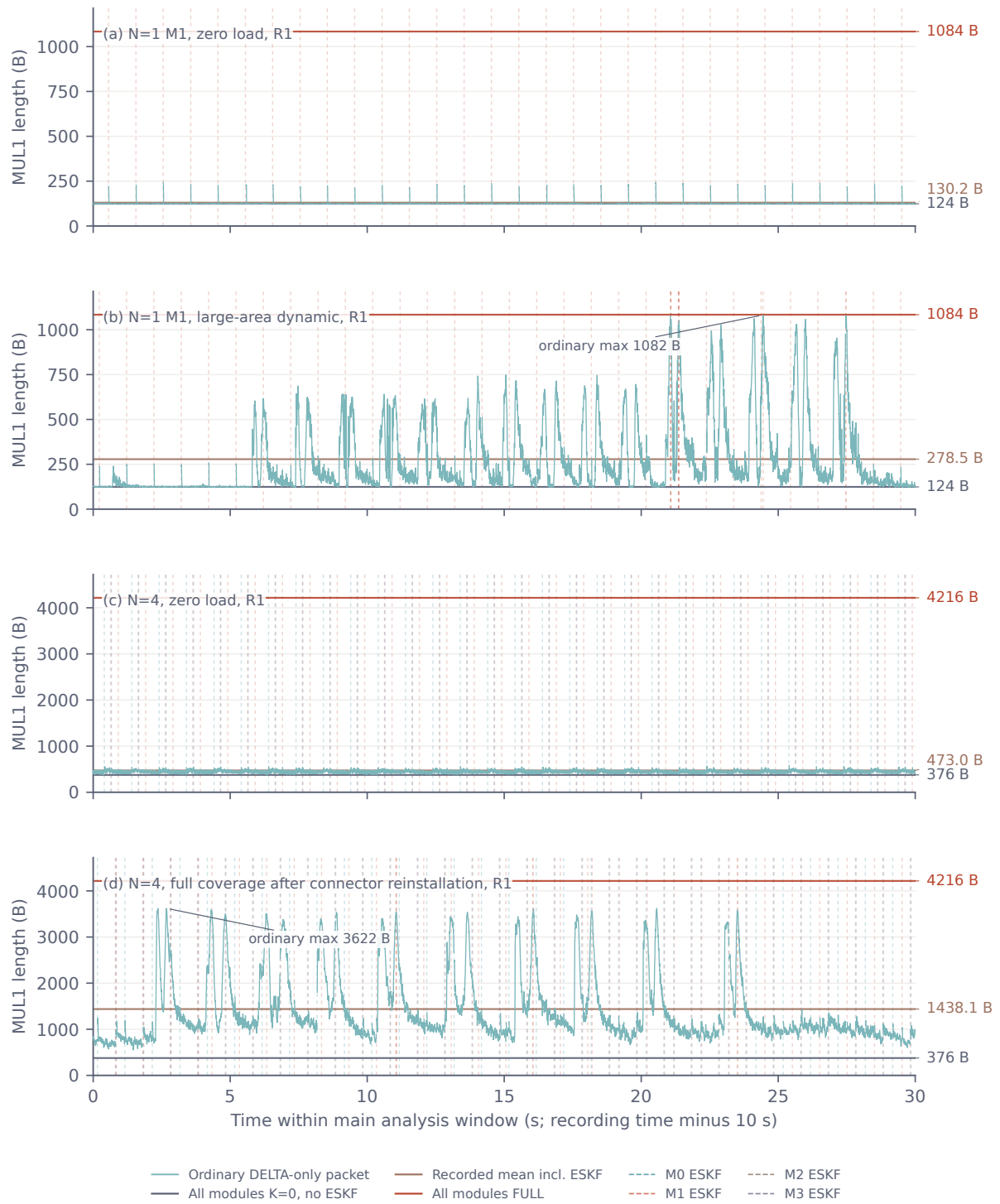


Figure C.1: Representative fixed-window traces for M1 and four-module zero and dynamic conditions. Dashed vertical lines mark recorded *ESKF* events; the four-module loaded trace is from the M0–M3 simultaneous 3/3 PASS batch.

Both diagnostic batches remain part of the results and limitations. The initial M1 batch contains inner sequence discontinuities despite CRC-valid traffic. In the pre-reinstallation $N = 4$ batch, R2 and R3 each have four unobserved outer sequence numbers and seven inner sequence-jump events; R2 also has one 40 B packet with no valid update from any target slot. The bridge intervals across the respective outer gaps are 522 and 453 ms. Outer missing numbers, inner sequence events and four absent slot updates are distinct quantities, not additive counts of physical scans lost. Subsequent *ESKF* frames restore the cache baseline but do not recover missing observations. The whole pre-reinstallation batch remains diagnostic. The operator identified poor connector contact and reinstalled the connector. In the subsequent batch, all three fixed main windows passed: 17,948 valid outer packets contained 71,792/71,792 expected module updates, with no CRC/layout, outer-sequence, inner-sequence, base-chain or post-anchor reconstruction finding. The three post-reinstallation windows provide the selected $N = 4$ dynamic statistics; they are consecutive temporal repeats in one setup rather than independent assemblies. Original zero-load labels in their acquisition metadata are preserved and corrected only in the derived index.

Chapter D

Power scalability: records and supplementary results

D.1 Data and reproducibility record

The continuous-read matrix combines 18 original YAML records with later supplements in `powerscalability/` `DATA/power_experiment_records/`. Within `transmission/`, seven additional voltage-only YAML files contain the four added two-module ZERO combinations, two added three-module ZERO combinations and the M0-only voltage retest. Four CSV files in `current_temperature_supplements/` retain the supplementary readings: `n2_zero_20260907.csv`, `n3_zero_20260907.csv`, `n2_max_20260907.csv` and `n3_max_20260907.csv`, with corresponding metadata JSON files. The ZERO CSVs supply current and temperature for the six added combinations; their voltages retain the separate voltage-record sources. The MAX CSVs include voltage, current and temperature for those combinations. Two later module retests in `targeted_retests/` provide the selected M0 ZERO current and M3 MAX voltage and current described in Table D.2. Five additional M0 current/voltage retests cover $N = 1$ under both loads and all M0-containing $N = 2$ combinations under ZERO. They are transcribed in `DATA/m0_n1_n2_retest_20260907/readings.csv`; `clarifications.json` records the confirmed N1 zero-load voltage of 3.252 V. The M0-M1 zero-load voltage photo was misassigned and is excluded; the operator confirmed 3.251 V for main/MIN/MAX instead. The revised view contains 30 configuration entries and 64 selected branch entries; a partial module retest is not counted as a complete configuration repeat.

The baseline analysis runs the following scripts in order, from the power directory. These scripts are in `script/Main/`:

```
summarize_voltage_review.py
summarize_n2_zero_20260907.py
summarize_zero_power_20260907.py
summarize_all_power_20260907.py
```

The preserved outputs in `DATA/all_power_review_20260907/` include configuration and branch tables, load contrasts, prediction checks, coverage, quality issues and a source manifest with SHA-256 hashes. Per-point provenance retains cross-record voltage assembly and relocated photograph paths. The subsequent `power_retest_revision.py` applies the two explicitly selected module updates and regenerates the derived tables in `DATA/report_v2_4_analysis/`. This versioned view preserves the baseline values and the retest sources, including which source voltages and other branch readings were retained.

The latest five updates are applied by `power_m0_preview_data_20260907.py`; the confirmed table snapshot is `DATA/report_v2_8_analysis/`. Its paired `load_voltage_drop_changes.csv` stores the 32 module-level contrasts with both source voltages, module voltages and drops. The report export uses `Final/tmp/power_v28_figures.py`, also preserved in the release archive. It retains the preview's data while exporting

vector figures without preview-status labels. The mean heatmap has been removed; the three-panel combination comparison is in the main results.

Report figure scripts are retained in `script/Main/`. Their revised inputs and provenance for this release are stored in `DATA/report_v2_8_figures/`. The module-count voltage-difference summaries retain individual values alongside means and sample SDs. Claim mappings are in the report’s `audit/` directory. Raw measurements and the earlier reproduction in `DATA/analysis_v2_1/` remain preserved.

The historical CSV, `DATA/power_raw.csv`, contains 13 blocked-standby rows. Eleven standby YAML copies in `no_transmission/` are excluded to avoid counting the same observations twice.

D.1.1 Continuous-read configurations

Table D.1 identifies all 15 continuous-read module combinations under ZERO and MAX. For the nine combinations marked O, each original YAML identifier is formed as `PS01_N{N}_F200_{L}_{S}_USB_RX_R1`, where L is the load label and S is the module string shown in the table. The six rows marked S are assembled from the supplements above and are not assigned invented original-run identifiers. All rows belong to the *FULL*, configured 200 Hz, continuous-read series. Each exact condition has one assembled entry, with no within-condition repeat SD. A dagger marks a branch-current sum containing a targeted retest and retained currents from the other branches.

Table D.1: All continuous-read module combinations and branch-current sums. O denotes original YAML records; S denotes supplementary source assemblies. Each module string and load label has one assembled entry. A dagger marks a partial retest update, not an independent new run.

N	Module string S	ZERO (mA)	MAX (mA)	Source
1	M0	18.018 [†]	19.675 [†]	O
1	M1	18.380	20.110	O
1	M2	18.360	19.897	O
1	M3	18.400	20.340	O
2	M0-M1	35.996 [†]	39.968	O
2	M0-M2	36.285 [†]	39.357	S
2	M0-M3	35.878 [†]	39.335	S
2	M1-M2	36.261	39.719	S
2	M1-M3	36.561	40.016	S
2	M2-M3	36.534	39.379	O
3	M0-M1-M2	54.564 [†]	59.328	O
3	M0-M1-M3	54.159	58.631	S
3	M0-M2-M3	54.130	58.279	S
3	M1-M2-M3	54.851	59.159 [†]	O
4	M0-M1-M2-M3	72.513	78.720	O

In the original $N = 1$ YAML files, the selected module’s readings are stored in `module0` fields irrespective of its label; the parser maps these fields using `module_ids`. Multi-module records use the explicit per-module fields. For legacy single-module rows with no branch-current value, the source-current reading is used. Module strings denote recorded assignments, not verified hardware UUIDs; prediction comparisons assume that assignment was retained.

For the latest five M0 updates, current MIN-MAX intervals are 18.00–18.02, 19.65–19.68, 17.92–17.99, 17.94–17.98 and 17.94–17.99 mA in the order of the first five rows. Their voltage MIN-MAX intervals are

Table D.2: Historical and selected module readings. Arrows show the earlier value followed by the value used in the revised analysis. The earlier three-module M0 voltage replacement predates its current retest; these columns do not imply simultaneous acquisition.

Configuration, load, module	Current (mA)	Voltage (V)	ΔV (mV)
M0			
ZERO, M0	18.160 → 18.018	3.235 → 3.252	45 → 28
M0			
MAX, M0	19.689 → 19.675	3.231 → 3.249	50 → 32
M0-M1			
ZERO, M0	17.961 → 17.941	3.234 → 3.251	47 → 30
M0-M2			
ZERO, M0	17.985 → 17.969	3.235 → 3.251	44 → 28
M0-M3			
ZERO, M0	17.985 → 17.956	3.237 → 3.250	42 → 29
M0-M1-M2			
ZERO, M0	17.968 → 18.051	3.224 → 3.241	52 → 35
M1-M2-M3			
MAX, M3	19.721 → 19.263	3.232 → 3.243	44 → 33

3.252–3.252, 3.249–3.250, 3.251–3.251, 3.251–3.251 and 3.250–3.250 V. The associated retained source main readings are 3.280, 3.281, 3.281, 3.279 and 3.279 V. Confirmation of a value or photograph assignment does not create an additional measurement; repeated file copies are not new observations.

The earlier three-module M0 current has instrument MIN/MAX readings of 18.00/18.08 mA. Its latest voltage confirmation supplies the main value 3.241 V only; the analysis retains the extrema from the earlier accepted voltage record, without presenting them as new measurements. The M3 retest has current MIN/MAX readings of 19.25/19.41 mA and voltage MIN/MAX readings of 3.243/3.244 V. The retained source main voltage is 3.276 V in both configurations, with its earlier provenance and extrema. For M3, the operator explicitly confirmed that this source value was unchanged; the new photograph is a module-end reading, not a new source measurement.

With the other branch currents retained, the zero-load sum for M0-M1-M2 changes from 54.481 to 54.564 mA. The loaded sum for M1-M2-M3 changes from 59.617 to 59.159 mA. Their source-output power estimates likewise combine different records. Neither update establishes a new simultaneous configuration measurement or an independent replicate.

D.1.2 Historical operating-state comparator

Table D.3 lists the historical CSV rows. All are *FULL*, R1 and labelled blocked standby without continuous USB reading. The 100 Hz rows remain historical screening observations. The operating-state figure shows 200 Hz single-module zero-load standby readings and zero-load and loaded continuous-read results. It does not treat separate sessions as a randomised paired intervention, nor identify an isolated causal effect of USB polling.

Table D.3: Historical blocked-standby identifiers, excluded from the main continuous-read curves. Each listed configuration has $n = 1$.

run_id
PS01_N1_F100_ZERO_M0_R1
PS01_N1_F200_ZERO_M0_R1
PS01_N1_F200_MAX_M0_R1
PS01_N1_F100_ZERO_M1_R1
PS01_N1_F200_ZERO_M1_R1
PS01_N1_F200_MAX_M1_R1
PS01_N1_F200_ZERO_M2_R1
PS01_N1_F200_ZERO_M3_R1
PS01_N2_F200_ZERO_R1
PS01_N2_F200_ZERO_M2_M3_R1
PS01_N3_F200_ZERO_R1
PS01_N3_F200_ZERO_M1_M2_M3_R1
PS01_N4_F200_ZERO_R1

D.1.3 Quality checks and unresolved metadata

Original records omit dates, start times and operator names. The analysis preserves photograph-filename date sets, including the original September 4–5 and supplementary September 6–7 batches, without substituting them for verified acquisition windows. Some supplementary zero-load currents and voltages come from different records. The nominal 30 s warm-up and 10 s measurement durations in the original forms are not independently confirmed. Default zero reset, current-limit, brownout and data-stop counters are not verified monitoring outcomes; supplementary photographs supply no equivalent fault history. The configured frequency and host-state labels likewise do not verify the actual clock, frame rate or uninterrupted operation during a reading.

Five historical continuous-read current main readings lie outside their recorded MIN/MAX interval by 0.002–0.005 mA. Four remain among the selected readings: the targeted M0 current retest lies inside its new interval and replaces one previously flagged main reading. All five original observations remain in the baseline record. The analysis flags values rather than clipping them or assigning a cause; display rounding and transcription should be checked against the source material. MIN/MAX readings are plotted directly where relevant and are not converted to SD. Measurement uncertainty cannot be reduced to these interval widths.

The selected view retains 64 branch-current and module-voltage MIN/MAX pairs and 30 source-voltage MIN/MAX pairs. These include retained earlier extrema where a targeted update did not supply new ones; they are not evidence of complete remeasurement. Two zero-load current minima required operator confirmation: M3 in M0–M3, 17.89 mA, and M0 in M0–M1–M3, 18.00 mA. The source photographs did not independently resolve those extrema; the metadata records the confirmation rather than implying image verification.

The M0 supply-connector event in M0–M1–M2, ZERO, is retained with its original 3.224 V main/MIN and 3.225 V MAX. That voltage point is replaced in the accepted view by the separate 3.241 V retest. The original source, M1 and M2 main voltages, 3.276, 3.250 and 3.246 V, were confirmed unaffected and retained. The resulting M0 difference of 35 mV is a user-confirmed cross-record estimate, not a simultaneous new measurement. The latest M0 branch-current retest is selected separately; other currents and all temperatures retain their original sources. The event and per-point provenance prevent either retest from becoming a new independent configuration or hiding the actual fault.

In the M3 MAX record the operator identified poor connector contact and reinserted the connector. The cur-

rent and module voltage changed as listed in Table D.2; equivalent loading between the two records was not confirmed. The 33 mV estimate uses the retained source reading confirmed unchanged, rather than a new simultaneous source measurement. Its earlier 44 mV value remains available in the baseline and historical comparison.

The nine original 2000 g MAX records and six supplementary MAX conditions without a mass value retain their different loading metadata. The 11 loaded prediction comparisons therefore describe nominal labels, not verified equivalent mechanical loading. Across both load labels, STM32 temperature point readings span 26.3–28.0 °C, Teensy readings 41.6–43.6 °C and regulator readings 28.7–31.2 °C. Entries labelled stable do not establish thermal equilibrium: ambient temperature, dwell time and a stability criterion are not verified. These are absolute point readings rather than temperature rises above a controlled ambient.

D.2 Supplementary observations

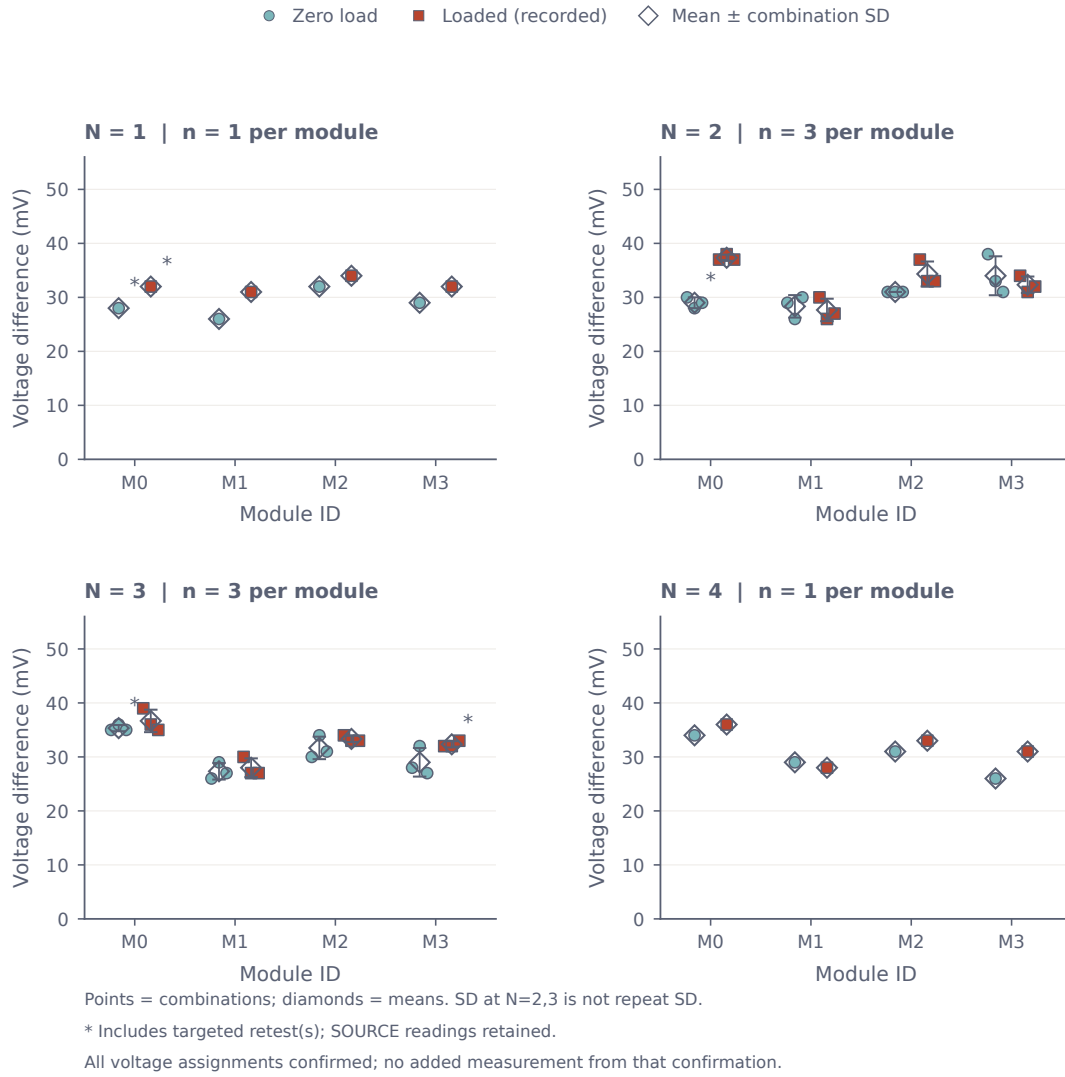


Figure D.1: Selected individual voltage differences and module-specific means for $N = 1, 2, 3, 4$. Each module contributes one, three, three and one configuration values, respectively. Error bars are sample SD across configurations at $N = 2$ and $N = 3$; no SD is estimated at $N = 1$ or $N = 4$. Overlapping module sets and the selected voltage retest estimates preclude interpreting these bars as independent repeatability estimates.

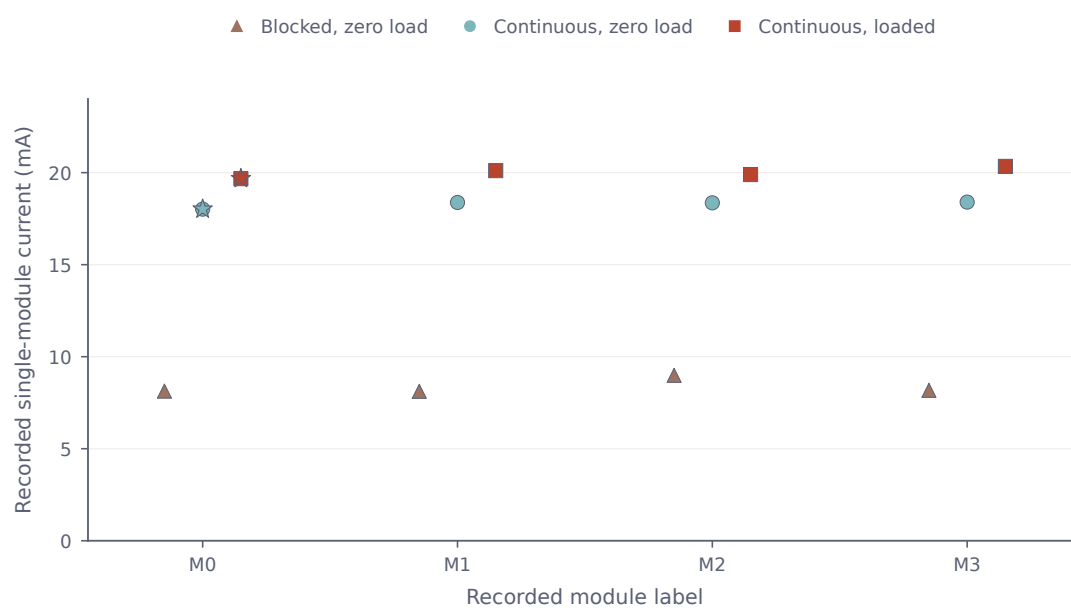


Figure D.2: Single-module readings at configured 200 Hz. Blocked standby is shown at zero load; continuous USB reading is shown at zero and recorded load. Each exact condition has $n = 1$; differences include session and setup effects.

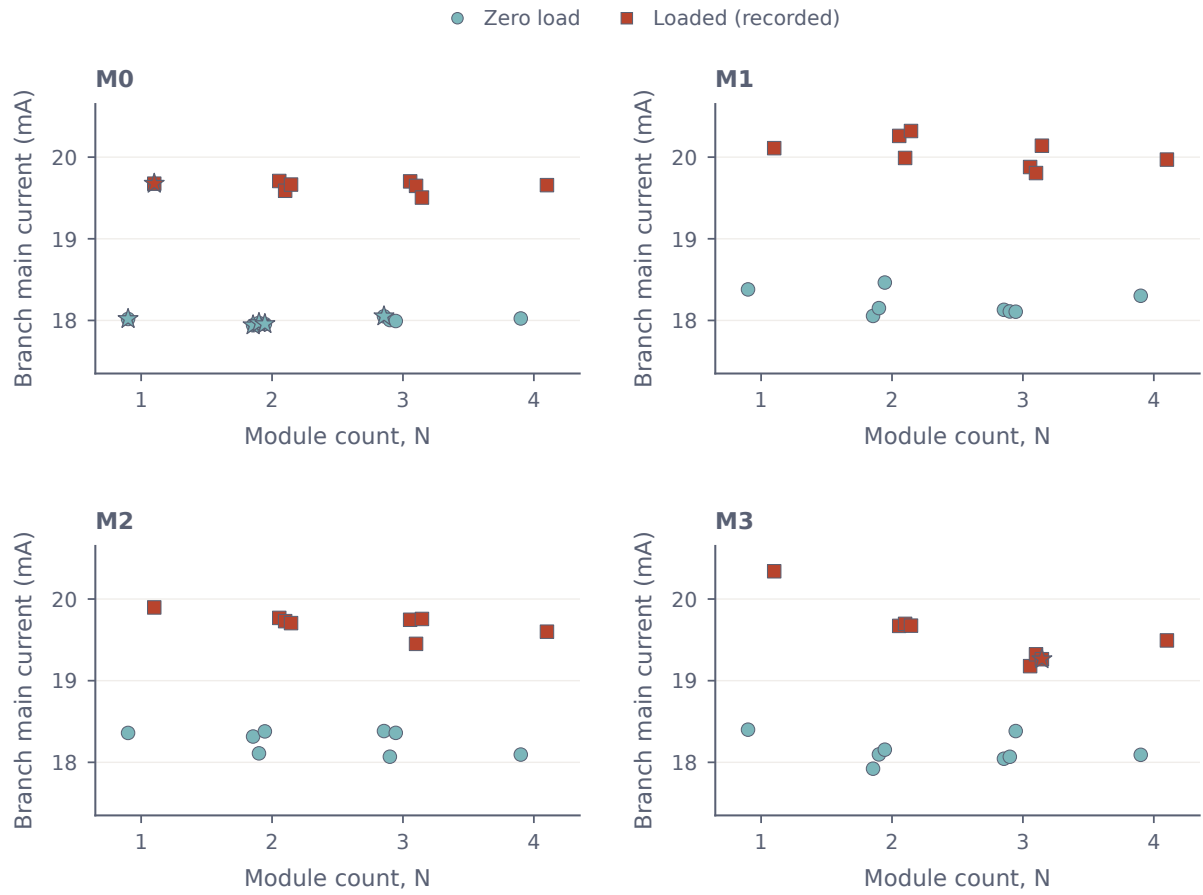


Figure D.3: Continuous-read branch current by record-labelled module, count and load across all 15 combinations. Points show the selected main readings, including seven targeted branch-current retests; they do not add complete configuration repeats. Module labels assume the physical assignment was unchanged.

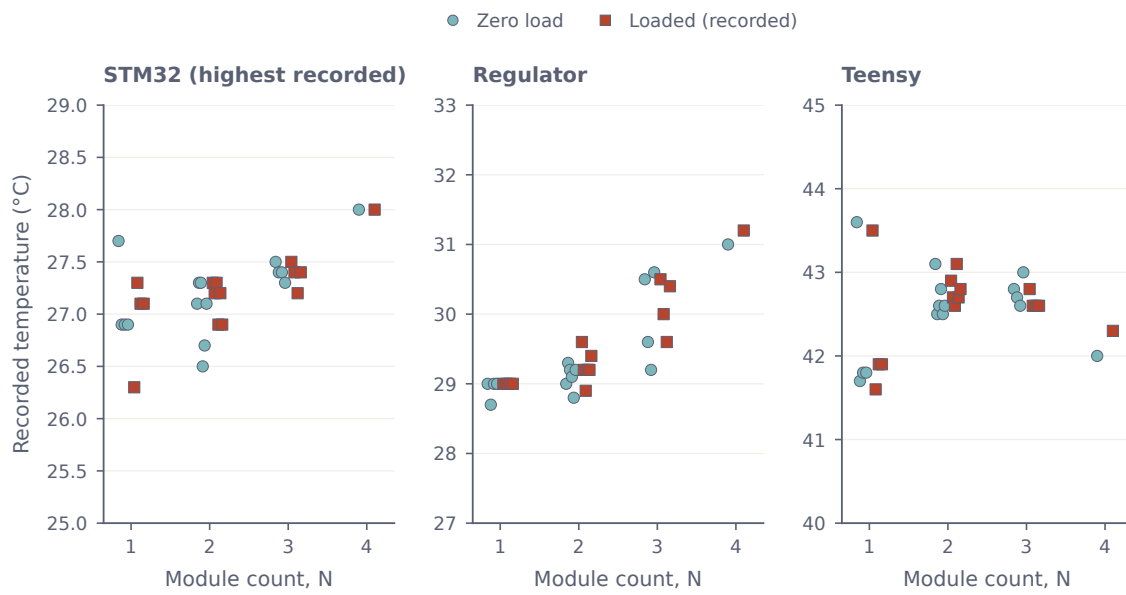


Figure D.4: Recorded continuous-read temperature observations for the highest available STM32 reading, regulator and Teensy. Each configuration has one set of point readings. Thermal equilibrium and matched dwell times are not verified; no repeatability or thermal-stability interval is inferred.

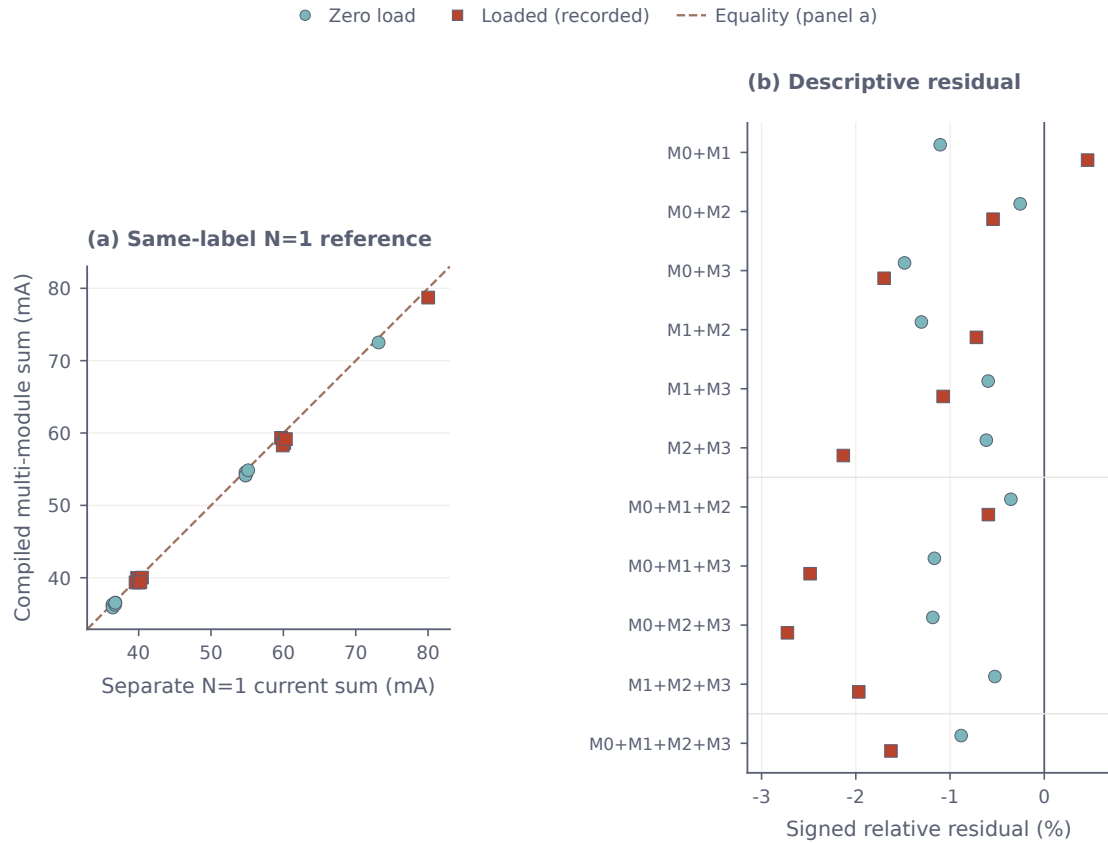


Figure D.5: Multi-module branch sums and signed residuals relative to separate single-module references, matched by module and nominal load label. Each label contains 11 comparisons; the residual denominator is the prediction. Comparisons share references and use one assembled entry per exact condition, including the seven partial retest updates. They are not independent repeats. Equivalent mechanical loading under MAX is unverified.

Chapter E

Calibration records and second-layer evaluation

E.1 Current source records and acquisition scope

The current results use the shared-sweep protocol `whole_layer_shared_endpoints_v3` in `docs/Final/Calibrationscalability/`. Both datasets were collected on 7 September 2026 from module M1. FSR2 provides the main-text results; FSR1 is reported below as a separate application of the same procedure to the other FSR layer. Each has its own 22-point sweep, 0 and 5000 g endpoints, 256 individual response models and one mean-response Shared LUT. Gamma is fixed at four in both cases.

Table E.1: Current M1 calibration datasets. Train phases counts complete layer-wide captures at 22 distinct masses spanning 0–5000 g. Captures and loads describe separate validation from 0 to 1969 g. Every training or validation capture contains 200 fresh FULL frames reduced by per-cell median. FSR2 has $n = 2$ validation captures at 140 g and $n = 1$ elsewhere; FSR1 has $n = 1$ at every validation mass. Session time is one training session from start to save, including manual loading and waiting, with no repeat SD available.

Layer	Train phases	Captures	Loads	Common cells	Session (s)
FSR2	22	21	20	256	334.1
FSR1	22	20	20	255	350.6

Under `DATA/module_1/`, training sweeps are stored in each layer’s `fit/raw/` directory:

- FSR2: `FSR2_pressure_sweep_20260907_100734_973698.json`.
- FSR1: `FSR1_pressure_sweep_20260907_082602_404396.json`.

Their zero and final maximum-load records supply the two-point endpoints without further acquisition. The FSR2 training masses, in grams, are 0, 73, 142, 215, 287, 359, 429, 496, 568, 639, 712, 778, 849, 921, 994, 1062, 1145, 1213, 1283, 1933, 3000 and 5000. FSR1 uses 0, 73, 143, 216, 288, 360, 430, 497, 570, 640, 713, 778, 850, 921, 994, 1061, 1145, 1213, 1282, 1932, 3000 and 5000.

Validation files are saved separately under each layer’s `evaluation/view_distribution/load_*/view_distribution_capture.json`. The FSR2 validation masses are 0, 70, 140 (twice), 212, 284, 352, 421, 489, 560, 630, 697, 800, 882, 966, 1035, 1107, 1179, 1249, 1319 and 1969 g. FSR1 uses the same distinct set except that 213 g replaces 212 g, with a single capture at every mass. A capture’s `models/` directory preserves the exact endpoint and fitted models used for its predictions. Model identities, source-sweep hashes and load ranges are checked before pooling; copied model files are matched by content, rather than by directory name.

E.2 Independent application to M1/FSR1

Figure E.1 shows the FSR1 model comparison from its own training sweep. This is not an application of the FSR2 LUT to FSR1. Of the 256 FSR1 cells, R6C15 has a reversed endpoint span of -160 ADC codes and a degenerate inverse containing only one knot. It is excluded from every method and capture, leaving a fixed common population of 255 cells. For the retained cells, positive endpoint spans range from 196 to 1473 ADC codes. Two zero-load and 20 upper-load instability flags are recorded across the layer; such noise flags alone do not cause exclusion.

M1 / FSR1: calibration functions ($\gamma = 4$)

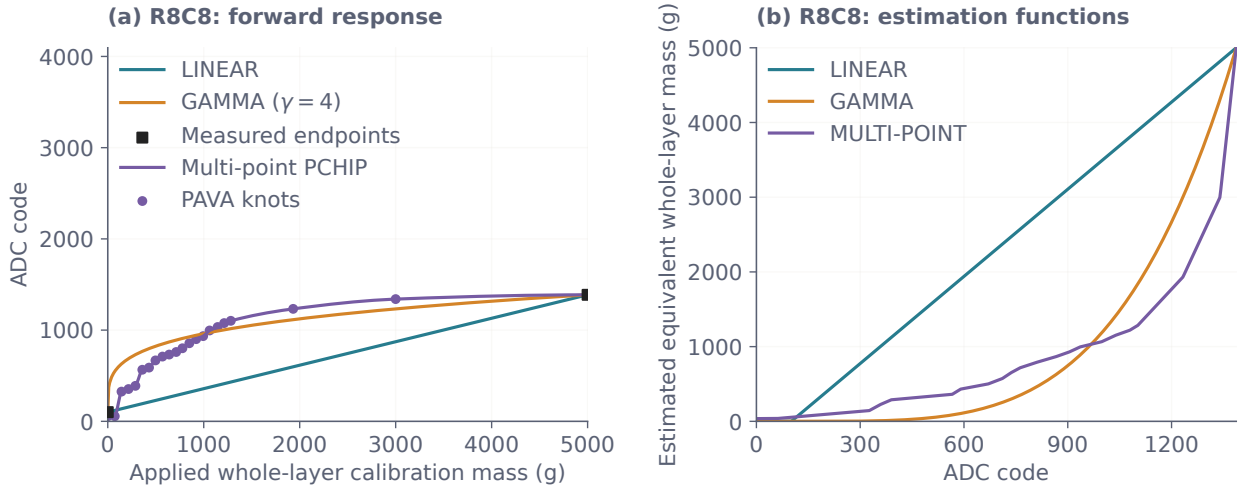


Figure E.1: M1/FSR1 training response and inverse mappings for representative cell R8C8, using this layer’s own models. Left: shared measured endpoints, PAVA-adjusted training knots and forward responses. Right: LINEAR, GAMMA ($\gamma = 4$) and FIT PRESS ADC–load mappings. All use the same 0–5000 g training sweep; each of its 22 points is a per-cell median of 200 FULL frames, with $n = 1$ capture per mass. The right-hand ADC axis ends at the cell’s measured upper-load reference. The comparison demonstrates application of the procedure to a second layer, not cross-layer transfer of fitted parameters.

Table E.2 summarises the separately recorded FSR1 validation. FIT PRESS has lower mean absolute layer bias and per-cell MAE than the Shared LUT, but the Shared LUT has a slightly lower mean-estimate APE (15.09% versus 15.57%). Thus, the FSR2 accuracy ordering is not universal even across these two layers on the same module.

At zero load, LINEAR, GAMMA, FIT PRESS and Shared LUT reconstruct 75.27, 1.97, 94.80 and 69.89 g, respectively. At 1969 g, FIT reconstructs 1833.78 g, with bias -135.22 g, per-cell MAE 385.23 g and spatial SD 566.09 g. The LINEAR spatial SD at this mass is smaller, 478.78 g, despite its much larger positive mean bias of 2419.86 g. This is a concrete example of why spatial spread and load accuracy require separate interpretation.

Table E.2: M1/FSR1 accuracy on a fixed 255-cell population. Per-cell LUT denotes FIT PRESS. Mean APE uses 19 non-zero captures; gram-valued means use all 20 captures, including zero. Per-cell MAE and RMSE are computed within each capture and then averaged, rather than pooling RMSE or taking RMSE of layer means. Each distinct load has $n = 1$ capture, so same-load repeat SD is unavailable. All methods use the same validation ADC, with independently acquired training models for FSR1.

Method	Mean APE (%)	Mean $ b $ (g)	Mean cell MAE (g)	Mean cell RMSE (g)
Linear	260.05	1694.65	1757.60	1957.64
Gamma ($\gamma = 4$)	54.33	409.71	693.89	935.29
Per-cell LUT	15.57	79.05	174.98	330.73
Shared LUT	15.09	93.99	382.47	644.03

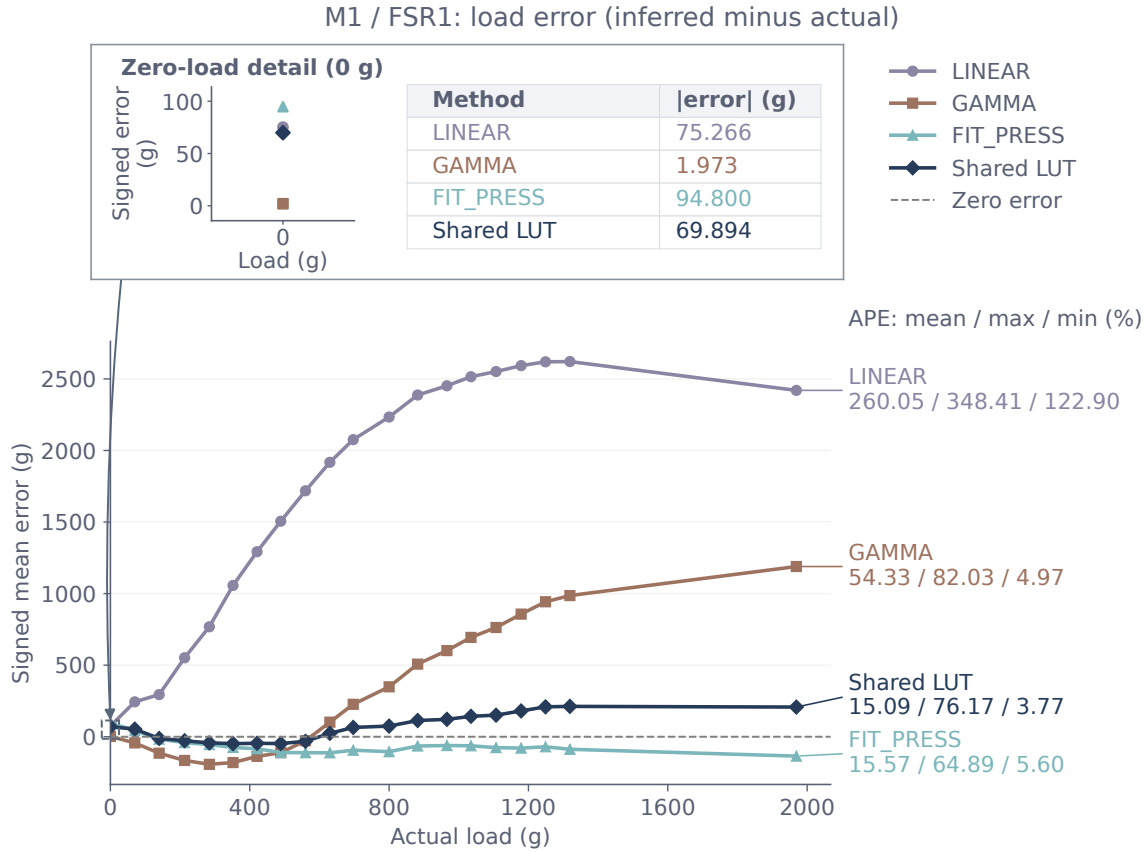


Figure E.2: M1/FSR1 signed error of the mean reconstructed mass for LINEAR, GAMMA, FIT PRESS and Shared LUT on the same 255 cells. Each of the 20 validation masses has $n = 1$ capture of 200 FULL frames; repeat SD is undefined and no error bar is drawn. Positive bias denotes overestimation. The zero-load detail reports bias in grams; zero is excluded from percentage errors only. Mean, maximum and minimum APE annotations use the 19 positive-load captures. The plotted 0–1969 g validation interval is narrower than the 0–5000 g calibration range.

E.2.1 FSR1 dispersion and model limitations

Across the 20 distinct FSR1 validation loads, FIT has lower variance than LINEAR at 19, lower IQR at 19, lower MAD at 18 and lower CV at five. Against GAMMA, the corresponding counts are 17, 15, 14 and 20. Figure E.3 retains the outcomes that favour the two-point methods as well as those that favour FIT.

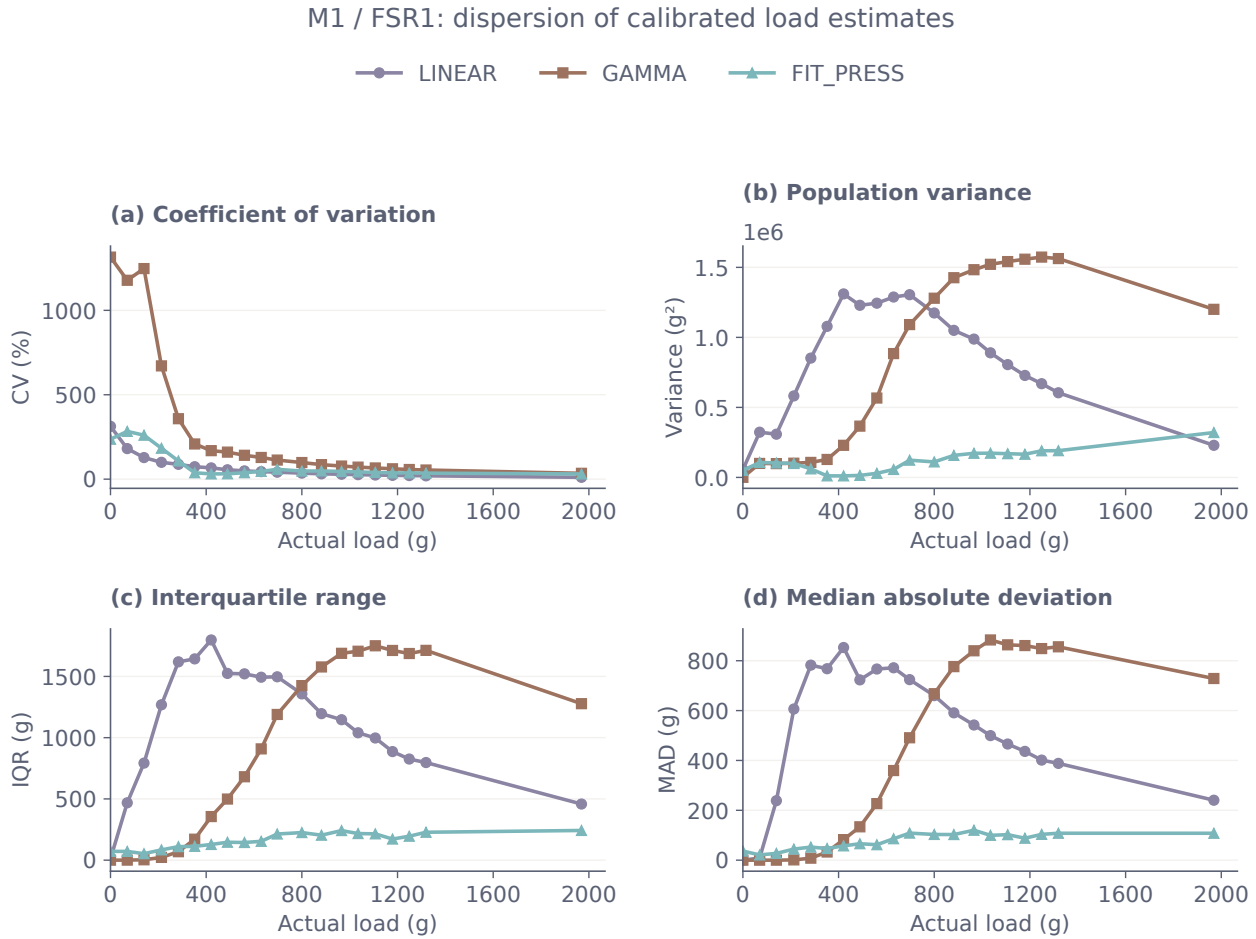


Figure E.3: M1/FSR1 CV (top left), population variance (top right), IQR (bottom left) and MAD (bottom right), computed directly from mass predictions on the same 255 cells. No Min–Max normalisation is applied. Spatial statistics use per-cell medians of 200 FULL frames. Each validation mass has $n = 1$ capture, so there are no repeat-SD error bars; spatial SD is not repeatability SD. CV is sensitive to small predicted means near zero. The validation axis covers 0–1969 g, while the models use the separate 0–5000 g calibration interval.

Among the retained 255 inverse tables, 243 contain merged plateau knots. There are 404 flat intervals, with median width 73 g and maximum width 4430 g. A plateau touches zero in 109 retained tables and the upper endpoint in 63. The mean lower-end inverse-LUT value is 34.15 g across these cells, which is a model property distinct from the measured zero-load reconstruction of 94.80 g. The excluded R6C15 table is counted separately as degenerate and is not allowed to contribute a constant prediction to the comparison.

The FSR1 training session lasts 350.6 s from start to save, including manual operations, and its acquisition windows total 23.14 s. There is one such session. Together with FSR2, this establishes that the same shared-acquisition procedure and analysis can be executed on both FSR layers with independent models. It does not establish between-module performance, repeatability after remounting, or reuse of a single layer’s LUT on the other layer.

E.3 Reproduction and preservation

From the report directory, `python tools/build_calibration_report_v27.py` checks the current raw captures, their temporal medians, frozen model provenance and evaluation summaries. The report evidence is stored in `audit/calibration_v2_7/evidence.json`, with source hashes and per-capture, per-load and plateau details. Generated numerical tables are in `audit/calibration_v2_7/tables/`; report figures are in `figures/calibration/v2_7/`. The figure generation uses the frozen calibration models without refitting them to validation data, and the audit verifies that the source data remain unchanged.

The calibration README documents the active GUI and separate commands for single-capture distributions, four-method load error and three-method dispersion. Batch analysis groups captures only when their module, layer, gamma, range and model identities agree. A shared LUT is recovered from the frozen training mean response, applied to each saved raw cell and evaluated on the existing common mask; it is not fitted using the validation targets. The stored audit also distinguishes capture-weighted headline errors from equal-weight distinct-load summaries.

The previous report version 2.6 is retained as the historical account. Historical acquisitions and their earlier normalisation policies are not pooled with the current September 7 datasets. Computational reproduction checks the consistency of saved observations and calculations; it does not supply the missing independent loading repeats, local pressure reference or cross-module experiment.

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